European Journal of Engineering Education

ISSN: 0304-3797 (Print) 1469-5898 (Online) Journal homepage: www.tandfonline.com/journals/ceee20

A systematic review of design thinking implementations in higher engineering education

Yuwei Deng, Claire Lucas & Wei Liu

To cite this article: Yuwei Deng, Claire Lucas & Wei Liu (09 Mar 2026): A systematic review of design thinking implementations in higher engineering education, European Journal of Engineering Education, DOI: 10.1080/03043797.2026.2640060

To link to this article: https://doi.org/10.1080/03043797.2026.2640060

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 09 Mar 2026.

Submit your article to this journal

Article views: 475

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ceee20

EUROPEAN JOURNAL OF ENGINEERING EDUCATION
https://doi.org/10.1080/03043797.2026.2640060

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

A systematic review of design thinking implementations in higher engineering education

Yuwei Deng, Claire Lucas and Wei Liu

Department of Engineering, King's College London, London, UK

ABSTRACT
Design thinking is widely adopted in engineering education to develop human-centred design capabilities, yet practices remain fragmented due to varied implementations, organisational barriers, and persistent gaps between theory and application. This literature review examines how design thinking has been conceptualised, implemented, and evaluated in higher engineering education from 2003 to 2023, analysing 76 empirical studies through a HOW (implementation strategies)–WHAT (core elements)–WHY (adoption rationales) framework. Findings reveal rapid growth after 2018, consolidation around three primary definitional sources, and widespread adoption of Stanford's five-stage model. Integration remains predominantly course-based rather than programme-wide. Four core competencies emerge across studies: creativity, collaboration, empathy, and problem-solving. Such consistency across institutional contexts and engineering disciplines underscores design thinking's potential to address persistent gaps in traditional curricula. Assessment approaches demonstrate methodological pluralism, though direct observation remains rare. This review offers evidence-based guidance for implementing design thinking in engineering education.

ABBREVIATIONS: STEM: Science, Technology, Engineering, and Mathematics; SDGs: Sustainable Development Goals; CDIO: Conceive, Design, Implement, Operate; PBL: Project-Based Learning; AI: Artificial intelligence

ARTICLE HISTORY
Received 19 March 2025
Accepted 27 February 2026

KEYWORDS
Design thinking; engineering education; design education; higher education

1. Introduction

1.1. The rise of design in engineering education

Design thinking, a human-centred approach to problem-solving, emphasises understanding user needs, iterative prototyping, and creative problem-solving (Brown and Katz 2009). This methodology integrates psychological insights, arrived at via empathy with users, into a non-linear process of experimentation and refinement (Brenner, Uebernickel, and Abrell 2016; Dell'Era et al. 2020; Kimbell 2011). In higher engineering education, design thinking has emerged as a critical strategy to equip students with the skills needed to navigate complex, real-world challenges (Dym et al. 2005). Design thinking can therefore be viewed as an additional, or even replacement, approach to traditional engineering component design which emphasises optimisation and analysis. This

CONTACT Wei Liu wei.liu@kcl.ac.uk

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

has taken place in parallel with a move away from waterfall design processes in industry towards a more agile approach. Scholars suggest it is best understood as a way of thinking that combines artistic creativity with scientific analysis to address problems comprehensively (Li et al. 2019). This shift responds to the long-standing criticism that traditional engineering curricula, with their emphasis on analytical approaches to well-defined problems ('science thinking'), often fail to prepare graduates for the ambiguous, open-ended nature of professional engineering practice where problems 'have no unique answers' (Berggren et al. 2003; E. F. Crawley et al., 2007; Samuel and Lewis 1991).

In addition to being an alternative way of describing the design process, within higher engineering education, design thinking has emerged as a critical pedagogical strategy to equip students with adaptable competencies for navigating rapidly evolving technological landscapes (Dym et al. 2005). For traditionally technical engineering disciplines, design thinking can provide a framework that enhances solution coherence while maintaining relevance to societal and industrial challenges (Dym et al. 2005). Design thinking, therefore, has the potential to play a central role in modern engineering curricula, since it fosters an interdisciplinary mindset that emphasises user-centred methodologies (Bosman 2019; McKilligan et al. 2017). By requiring empathy and iterative engagement with end-user perspectives, it cultivates pragmatic problem-solving skills suited for real-world complexities and consideration of societal outcomes (Koh et al. 2015; Scheer, Noweski, and Meinel 2012). The design thinking approach also scaffolds abstract concepts such as creativity and innovation into teachable, structured processes, enabling students to bridge theoretical knowledge with practical application (DeHaan 2009).

Several systematic reviews have examined aspects of design education in engineering, providing important context for the present study. For instance, Van Helden et al. (2023) conducted a systematic review of collaborative learning in engineering design education, identifying key factors that promote effective teamwork in design contexts. Their review highlights collaboration as a critical competency in design practice, examining pedagogical strategies for fostering teamwork, communication, and shared problem-solving. While collaboration often emerges as an element within design thinking approaches, Helden et al.'s focus differs from an investigation of design thinking as a comprehensive pedagogical approach. Other reviews, such as de Paula, Cormican, and Dobrigkeit (2021), have tracked the contribution of design thinking to new product development from a management perspective, examining its impact in business and innovation contexts.

Our review addresses this gap and complements existing work by focusing specifically on the 'design thinking' movement in higher engineering education. We employ a comprehensive HOW-WHAT-WHY framework to synthesise evidence on its conceptualisations (what design thinking means in engineering education), implementations (how it is taught and integrated), and the rationale behind its adoption (why engineering educators are incorporating these approaches).

1.2.1 Existing reviews: scope and limitations

Several systematic reviews have examined aspects relevant to design thinking in engineering contexts, but analysis reveals three critical limitations that collectively necessitate the present study.

The most methodologically rigorous reviews on design thinking focus on contexts other than higher engineering education. Yu, Yu, and Lin (2024) found positive effects of design thinking on student learning but did not distinguish engineering from broader STEM contexts, and their geographical distribution raised questions about global generalisability. Li and Zhan (2022) reviewed design thinking but explicitly excluded higher education, focusing solely on K-12 contexts. Saseendran and Thomas (2025) documented recent publication surges but focused primarily on K-12 and science rather than engineering specifically. This pattern reveals that rigorous design thinking reviews systematically exclude or fail to distinguish the higher engineering education context that has seen the most rapid adoption of design thinking frameworks.

The only systematic review specifically targeting design thinking in engineering education (Acebo, Miguel-Dávila, and Herrera 2021) covered literature only through 2018, finding that design thinking promotes transversal skills but identifies a fundamental lack of standardisation.

Numerous rigorous reviews examine related pedagogical approaches but treat design thinking peripherally rather than as a Sukackė et al. (2022) study on PBL/PjBL/CBL, O'Connor, Power, and Blom (2023) study on CDIO, all incorporate design elements but do not explicitly examine design thinking as distinct methodology or map relationships between design thinking and these frameworks. Van Helden et al. (2023) reviewed collaborative learning in engineering design education. Reviews on creativity and innovation (Aguilera and Ortiz-Revilla 2021) similarly lack clear conceptual frameworks for design thinking specifically. This implicit treatment leaves unresolved questions about how design thinking relates to, differs from, or complements established engineering education approaches.

While previous systematic reviews have examined specific aspects of design education, including collaboration, creativity, pedagogical frameworks, and assessment methods, or investigated design thinking's application in business management, K-12 education, and broader STEM contexts, none have specifically focused on how 'design thinking' as a distinct pedagogical movement has been conceptualised and implemented across higher engineering education. Given the increasing adoption of specific 'design thinking' frameworks and terminology in engineering programmes worldwide, understanding the state of knowledge specifically about this movement, its definitions, implementations, outcomes, and challenges, represents a significant gap in the literature.

1.2.2 Justification for the present review

The identified limitations collectively justify a comprehensive systematic review specifically on 'design thinking' as a distinct pedagogical movement in higher engineering education. First, design thinking has emerged as a movement with specific mindsets, processes (e.g. Stanford d.school's empathise-define-ideate-prototype-test), and pedagogical philosophies that distinguish it from generic engineering design. Second, the post-2018 period represents critical growth requiring updated synthesis beyond existing reviews' coverage. Third, engineering education contexts possess unique characteristics – professional competencies, accreditation requirements, established frameworks (PBL, CDIO), and implementation contexts (labs, studios, capstones) – that distinguish them from K-12 or broader STEM education. Fourth, relationships between design thinking and established engineering pedagogies remain underexplored. Fifth, consistent gaps across reviews indicate educators lack clear, evidence-based implementation guidance specific to engineering contexts. This review addresses these gaps through comprehensive, systematic synthesis of design thinking conceptualisations, implementations, and outcomes in higher engineering education.

1.3. Defining key terms: design, thinking, and design thinking

Definitions and emphases of design thinking vary across contexts (Golkar 2020), and when incorporated as a curriculum framework, these variations lead to distinctive approaches to its application in academic settings (Dorst 2011). To clarify our scope and inform our inclusion criteria, we provide working definitions of the key terms central to this review.

Design: Following Skolimowski (1966), we understand design in engineering as creating 'a reality according to our designs' rather than investigating given reality as in science. Design involves intentional planning and decision-making to bring new artfacts, systems, or solutions into existence. Crucially, design problems are characterised by their open-ended nature – they 'have no unique answers' (Samuel and Lewis 1991) and require synthesis rather than purely analytical approaches (Dym et al. 2005). Design is therefore fundamentally different from problem-solving in well-structured scientific domains.

Thinking: Our conceptualisation of 'thinking' acknowledges multiple theoretical perspectives that inform how design thinking is understood and taught. While cognitivist frameworks locate thinking primarily in internal mental processes, reasoning strategies, problem representations, and decision-making heuristics (Eastman, Newstetter, and McCracken 2001; Oxman 2001), pragmatist and post-cognitive perspectives (Mitcham 1994; Schön 2017) emphasise that thinking occurs through

action, interaction with materials, and use of tools and representations. As Mitcham (1994) argues, engineers 'think through technology' – thinking is distributed across minds, bodies, tools, and social contexts (Brereton 2004). In design thinking, both dimensions are relevant: the cognitive strategies and mindsets designers employ, and the material and social practices through which thinking is enacted.

Design Thinking: For this review, we conceptualise 'design thinking' as it has been operationalised in engineering education contexts – specifically referring to pedagogical approaches that typically:

1. Emphasise human-centred problem-solving grounded in understanding user needs, experiences, and contexts
2. Follow iterative, non-linear processes involving stages such as empathising, defining problems, ideating, prototyping, and testing
3. Value abductive reasoning and comfort with ambiguity alongside analytical thinking
4. Promote interdisciplinary collaboration and integration of diverse perspectives
5. Encourage rapid prototyping and learning through making as modes of inquiry
6. Are explicitly identified as 'design thinking' by practitioners and researchers, often drawing from frameworks popularised by Stanford's d.school and IDEO

This conceptualisation distinguishes design thinking from traditional engineering design methodologies that may emphasise more linear, optimisation-focused, and analysis-heavy approaches. It also distinguishes design thinking from general design education that may not adopt the specific process models, terminology, and pedagogical commitments of the design thinking movement. We acknowledge that 'design thinking' has diverse definitions and implementations across contexts; our review investigates this diversity by systematically examining how the concept has been understood, adapted, and applied in different engineering education settings.

Scope limitation and acknowledgment: By focusing specifically on literature that explicitly uses the term 'design thinking', this review necessarily excludes important research on related concepts such as design cognition, engineering design processes, designerly ways of knowing, and general engineering design education. While this narrower scope limits our ability to capture the full breadth of design-related pedagogical approaches in engineering education, it enables us to examine 'design thinking' as a distinct educational movement with specific characteristics, frameworks, and discourses. This deliberate choice reflects our research aim to understand how this pedagogical approach has been conceptualised and implemented, rather than to comprehensively review all design-related education in engineering. We acknowledge this as a limitation of our search strategy and caution against overgeneralising our findings to design education more broadly. Future research employing broader search strategies could provide complementary insights into the wider landscape of design pedagogy in engineering education.

Given the complexity of design challenges and the different perspectives on their solutions, it is crucial to have a comprehensive understanding of the various practices and conceptualisations of design thinking in higher engineering education. This review evaluates the research landscape of design thinking within higher engineering education, highlighting its multifaceted influence and the dynamics of its integration across engineering disciplines.

1.4. Study objectives

Through a detailed systematic analysis of empirical studies, this paper identifies key trends and characteristics of design thinking research in engineering education, providing deeper insights into its implementation and the driving forces behind its adoption in higher engineering education contexts. The research questions guiding this study are:

RQ1: What are the characteristics and trends of research about design thinking in higher engineering education?

RQ2: How is design thinking currently taught and implemented in higher engineering education, including implementation strategies (how), the core elements of design thinking (what), and the rationale for incorporating design thinking (why)?

The core elements examined in RQ2 include the modes of thought, objects of thought, capacities, actions, processes, and artefacts associated with design thinking pedagogy and practice in engineering education contexts.

2. Background

2.1. Conceptual definition and research scope

2.1.1 'Design thinking' as a specific educational movement

In the field of engineering education, the concept of 'design thinking' requires clear definition at the outset. While broader literature on design cognition, engineering design processes, and designerly ways of knowing (Archer 1979; Cross 1982; Kruger and Cross 2006) have made important contributions to our understanding of how designers think and act, 'design thinking' has evolved over the past two decades into a distinct pedagogical approach within engineering education. This distinction is crucial for understanding our research focus.

Beginning in the early twenty-first century, 'design thinking' was broadly promoted as a specific methodology through the work of IDEO and Stanford's d.school (Brown 2008). This approach emphasises practices, building empathy through user research, rapid ideation, prototyping, and iterative testing, organised into distinctive process models (e.g. empathise-define-ideate-prototype-test). While these practices stem from professional design traditions, 'design thinking' as taught in engineering education typically presents them as a coherent, transferable methodology applicable across different problem domains.

2.1.2 Distinctions from related concepts

To accurately understand the uniqueness of 'design thinking', it is necessary to distinguish it from related but different concepts: Design cognition (Brereton 2004; Eastman, Newstetter, and McCracken 2001; Oxman 2001) primarily studies cognitive processes, mental representations, and problem-solving strategies in design activities. This research tradition leans toward the cognitive level, typically based on protocol studies, cognitive task analysis, and cognitive science frameworks. While helpful for understanding how designers think, design cognition research usually focuses on describing cognitive phenomena rather than prescribing pedagogical frameworks.

Design processes and heuristics (Atman et al. 2007; Daly, Adams, and Bodner 2012) examine the procedural aspects, strategies, and rules of thumb that designers use. This literature typically focuses on describing and comparing expert versus novice design behaviours, identifying effective strategies, and understanding decision-making patterns. While overlapping with design thinking, this research tradition usually emphasises descriptive analysis of design practice rather than the prescriptive pedagogical frameworks characteristic of design thinking approaches.

Designerly ways of knowing (Cross 1982) explores the epistemological foundations of design – the forms of knowledge, modes of inquiry, and ways of understanding the world that are unique to the design discipline. This philosophical and theoretical tradition studies design as a fundamental human capability and intellectual discipline, providing important conceptual foundations but typically operating at an abstract level beyond specific pedagogical practices.

Engineering design education (Carberry and McKenna 2014; Dym et al. 2005; Juhl and Lindegaard 2013) encompasses the broad field of teaching design in engineering contexts, including traditional engineering design methods, capstone projects, design courses, and various pedagogical approaches. 'Design thinking' represents a specific pedagogical approach within this broader field, characterised by its emphasis on human-centeredness, empathy, and iterative prototyping.

2.1.3 Defining the scope of this study

Based on the above distinctions, this study specifically focuses on 'design thinking' for several reasons: First, it represents a distinct discourse community, educators, researchers, and institutions explicitly identify their approaches as 'design thinking', indicating participation in a specific pedagogical movement with shared assumptions and practices.

Second, it embodies specific pedagogical commitments, design thinking approaches typically emphasise human-centeredness, empathy, tolerance for ambiguity, and interdisciplinary collaboration in ways that distinguish them from traditional engineering design education (Dym et al. 2005). Third, it has practical implications for curricular decisions – engineering programmes increasingly consider whether and how to adopt 'design thinking' frameworks. Understanding how this specific approach has been conceptualised and implemented provides actionable insights for such decisions. Finally, it reflects an identifiable educational reform movement – similar to how CDIO (E. F. Crawley et al. 2007) and Problem-Based Learning (PBL) represent specific educational reforms, 'design thinking' has become a distinct movement with its own advocates, practices, and institutional implementations. Therefore, our review studies 'design thinking' as a specific pedagogical phenomenon in engineering education, acknowledging that this narrower focus necessarily excludes important perspectives from the broader design cognition and engineering design literature. We draw upon these broader literatures to contextualise and frame our findings while maintaining focus on how 'design thinking' is understood and practiced in engineering education contexts.

2.2. Theoretical foundations

2.2.1 Two competing theoretical perspectives

Having established the conceptual boundaries of design thinking, understanding its theoretical roots becomes crucial. Understanding design thinking in engineering education requires deep exploration of two influential but competing theoretical perspectives: those of Herbert Simon and Donald Schön. These perspectives represent fundamentally different orientations toward the nature of design and how it should be understood and taught, and this theoretical divide continues to profoundly influence design thinking educational practices today.

Simon (2013) was one of the early pioneers in viewing design as a science or way of thinking since 1969, conceptualising design as 'the science of the artificial' in his book of the same name. He emphasised systematic problem-solving processes, search strategies, and satisficing (seeking satisfactory rather than optimal solutions). This perspective views design as primarily a cognitive, rational activity that can be analysed and taught through structured methods. Simon's approach emphasises well-structured problems amenable to systematic search and the application of explicit principles and methods.

Donald Schön's reflective practice perspective. In sharp contrast, Schön (1992) proposed a different perspective based on extensive research with design professionals. He argued that effective design involves 'reflection-in-action' – a form of knowing-in-practice and professional artistry in which practitioners engage in an ongoing 'conversation with the materials of a design situation' (Schön 1992, 131). From this perspective, design is not merely about applying analytical frameworks but involves improvisation, tacit knowledge, and the ability to frame and reframe problems in action.

2.2.2 Core differences in the theoretical divide

The differences between these two perspectives have enduring significance for design thinking education, manifested in three key areas: the nature of problems: Simon (1973) emphasises well-structured problems amenable to systematic analysis, while Schön (1987) focuses on 'the indeterminate zones of practice', where problems must be framed rather than simply solved. Simon's approach suggests design can be taught through explicit principles and methods, while Schön emphasises

tacit knowing, professional artistry, and learning through reflective practice in authentic contexts. Simon's framework tends to separate problem analysis from solution generation, whereas Schön's 'reflection-in-action' integrates thinking and doing through prototyping, experimentation, and dialogue with materials.

2.2.3 Different conceptualisations of 'thinking'

The complexity of theoretical foundations extends beyond the Simon-Schön opposition. Our understanding of design thinking necessarily depends on how we conceptualise 'thinking' itself. Different theoretical perspectives inform this conceptualisation, with significant implications for pedagogy and research.

The cognitivist perspective (Eastman, Newstetter, and McCracken 2001; Oxman 2001) conceptualises thinking primarily as internal mental processes – problem representation, reasoning strategies, memory retrieval, and decision-making heuristics. From this perspective, design thinking involves specific cognitive strategies such as abductive reasoning, pattern recognition, and mental simulation. The pragmatist perspective offers a contrasting view. Technology pragmatist philosophers, particularly Mitcham (1994), argue that 'design constitutes the essence of engineering' and, crucially, that engineers 'think through technology'. From this perspective, thinking is not solely or primarily located in the mind – it occurs through material engagement, tool use, and technological practice.

The post-cognitivist perspective further extends this view, seeing thinking as fundamentally distributed across minds, bodies, tools, social contexts, and material environments. Contemporary theories, including distributed cognition (Brereton 2004),embodied cognition, situated cognition, and activity theory emphasise that cognition is not confined to individual minds but is achieved through the coordinated use of material and social resources.

2.2.4 Implications of theoretical perspectives for pedagogical practice

These different conceptualisations lead to different emphases in design thinking education. If adopting a cognitivist perspective, educational focus centres on developing individual cognitive skills such as visualisation, analogical reasoning, and problem reframing. In contrast, a pragmatist perspective emphasises learning through making, material experimentation, and prototype development, seeing thinking as emerging through material engagement. A post-cognitivist perspective requires creating learning environments that support distributed cognition, integrating tools, social interactions, and material resources as constituent parts of the thinking process. As Bucciarelli (2001) notes, echoing Schön's influence: 'We cannot teach design the way we teach science'. These foundational perspectives, Simon's structured rationality versus Schön's reflective practice, remain visible in contemporary design thinking education (Owen 2007). Some approaches emphasise structured process models and analytical tools (more aligned with Simon's view), while others emphasise empathy, experimentation, tolerance for ambiguity, and learning through making (more aligned with Schön's view).

2.3. Historical development

Different understandings of theoretical foundations have directly influenced the historical development trajectory of design thinking in engineering education. This development can be divided into three major phases, each reflecting the ongoing integration of theory and practice.

2.3.1 Early exploration phase (1957-1990s)

The early development of design thinking in education began with institutional-level innovations. In 1957, Professor Arnold at MIT offered one of the first courses titled 'Human-Centred Design and Engineering', marking an early integration of design thinking principles in academia (Von Thienen et al. 2018). This period was characterised by early interdisciplinary courses combining engineering with scientific principles, which played important roles in shaping design thinking's place in

engineering education (Kolodner 2002; Kolodner, Gray, and Fasse 2003). This phase was primarily exploratory, with educators attempting to translate the tacit knowledge of design practice into teachable content, though without a unified methodological framework.

2.3.2 Methodological formation phase (1990s–2005)

The 1990s marked a turning point, with Tim Brown and his colleagues at IDEO particularly articulating the concept of 'design thinking' (Nussbaum 2004). Although this concept has deeper historical roots in the evolution of design as an academic discipline, IDEO's contribution was to systematise it into a replicable methodology. Another important milestone in this period was the founding of Olin College of Engineering in 1997, which recognised design thinking as a key skill for cultivating innovative engineers (Somerville et al. 2005). This represented institutional-level recognition of design thinking's importance, viewing it not as a supplementary skill but as a core component of engineering education.

2.3.3 Institutionalisation and dissemination phase (2005–present)

In 2005, Stanford University formally began teaching design thinking to engineering students through its d.school (Plattner 2007), further consolidating design thinking's significance in higher education curricula. The d.school's approach particularly emphasised empathy, rapid prototyping, and interdisciplinary collaboration, elements that have now become hallmarks of design thinking education. This phase has witnessed design thinking's evolution from a novel approach into a mainstream pedagogical framework in engineering education, adopted and adapted globally.

2.4. Educational dimensions of design thinking

The trajectory of historical development shows that design thinking in engineering education has evolved three interrelated but distinct dimensions (Brenner, Uebernickel, and Abrell 2016): as a process (Grönman and Lindfors 2021), as competencies (Grönman and Lindfors 2021; Razzouk and Shute 2012), and as a pedagogical framework (Goudarz Poursharif et al. 2021). These three dimensions together constitute the complete picture of contemporary design thinking education.

2.4.1 As a process: methodological framework

Design thinking as a process emphasises an iterative, human-centred approach to problem-solving. Typical process models include five stages: empathise (understand user needs), define (clarify the problem), ideate (generate ideas), prototype (build solutions), and test (evaluate and improve). This process view emphasises that design is not linear but cyclical and iterative, allowing continuous refinement based on user feedback and testing results.

This process-oriented approach provides students with concrete operational steps, making the abstract innovation process actionable and learnable. However, over-reliance on fixed processes may also limit the flexibility of creative thinking.

2.4.2 As competencies: core skills development

Design thinking as competencies is recognised as a key theme in engineering education curriculum objectives, based on the design thinking competency model proposed by Razzouk and Shute (2012). Facing the complexity and uncertainty of our changing times, engineers increasingly need the innovation support of design thinking to assist with problem identification, solution, and subsequent system design.

Research shows that incorporating design thinking into engineering education produces several positive outcomes. First, it promotes the development of comprehensive skills and innovative thinking by emphasising interdisciplinary, holistic, and problem-solving abilities (Bosman 2019). Second, it enhances students' creativity, sustainability awareness, and career readiness (Crawley et al. 2007). Additionally, research shows design thinking can improve students' leadership, algorithmic thinking,

entrepreneurship, and critical thinking skills, cultivating a culture of design and creation (Henriksen, Mehta, and Mehta 2019).

More importantly, design thinking complements the shortcomings of traditional engineering education. Traditionally, engineering education primarily focuses on critical thinking within technical domains, developing students' proficiency in managing tasks (Berggren et al. 2003). In contrast, design thinking aims to generate ideas and explore new possibilities, providing concrete suggestions for conveying complex phenomena that are not overly abstract, understandable to students, and implementable by teachers (McLaughlin et al. 2019).

2.4.3 As a pedagogical framework: integration with existing approaches

The third dimension of design thinking in engineering education is as a pedagogical method that can be integrated with existing instructional frameworks. Two prominent examples illustrate this integration: Integration with Problem-Based Learning (PBL). Problem-Based Learning as a pedagogical model places students at the forefront and redefines the teacher's role (Kuo, Tseng, and Yang 2019; Goudarz Poursharif et al. 2021). Kuo et al. (2022) reported positive outcomes of PBL in engineering education, demonstrating its effectiveness for both students and teachers. In engineering education, PBL has become one of the most used pedagogical approaches for promoting design thinking. Research (Jiang and Pang 2023; Mills and Treagust 2003) reveal positive correlations between students' design experiences and their intellectual development, demonstrating PBL's positive impact on innovation, professional skills, and practical abilities.

Application of the CDIO Framework. The Conceive, Design, Implement, Operate (CDIO) framework is another engineering education pedagogical approach that incorporates design thinking elements (Crawley et al. 2007). Through hands-on projects and practical learning, CDIO integration in engineering education aims to encourage students to apply theoretical knowledge to real-world situations and develop active learning and problem-solving abilities.

This approach often involves interdisciplinary collaboration, helping students develop communication and teamwork skills. Furthermore, CDIO meets industry needs by presenting students with real-world engineering problems, enhancing their career readiness. Several engineering institutions have adopted the CDIO pedagogical approach, recognising its ability to bridge the gap between theoretical knowledge and practical application (Edström and Kolmos 2014).

3. Methods

In response to the research questions, a systematic literature review was conducted following the methodologies outlined by Borrego, Foster, and Froyd (2014), Davies (2000), and Petticrew and Roberts (2008). The process involved Search, sift, and select based on established search and screening criteria (Gough, Thomas, and Oliver 2017). Once the relevant studies were identified, the analysis proceeded with data extraction and coding to critically evaluate the findings. These codes were subsequently synthesised for interpretation and reporting. The review concluded with an assessment of the methods' validity and reliability, highlighting any potential limitations to ensure the integrity of the findings.

3.1. Search, sift and select

3.1.1 Search strategy

The initial search using Boolean phrases commenced on 25 June 2023, and concluded on 19 August 2023. Given our primary objective to understand the implementation practices of design thinking in higher engineering education and the associated developmental outcomes, consultations were held with subject librarians to determine the appropriate databases. Scopus and Web of Science were selected as they are among the most comprehensive global databases for engineering literature. ERIC was chosen for its extensive and highly regarded collection of educational literature.

In this study, STEM (science, technology, engineering, and mathematics) serves as the foundational framework for our research approach (Sanders 2008). While engineering remains central, the broader STEM perspective enhances understanding through its interdisciplinary nature, allowing students and researchers to integrate diverse knowledge and methods for more effective problem-solving. With assistance from subject librarians, we developed database-specific search strings, detailed in Table 1.

3.1.2 Inclusion and exclusion criteria

To be included in this review, articles needed to meet the following five inclusion criteria (IC):

Publication Year and Language (IC1): Restricted by database search, studies must be published in English between 2003 and 2023. This timeframe is chosen based on significant developments in design thinking, which expanded beyond its traditional domains in engineering and the arts during this period (Mabogunje, Sonalkar, and Leifer 2016). The selection aligns with periods used in other design thinking reviews, ensuring relevance and comparability (de Paula, Cormican, and Dobrigkeit 2021).

Types of Publications (IC2): The inclusion criteria specify that eligible studies must be published in archival journals or peer-reviewed conference proceedings. This ensures that the content has undergone peer-review processes. Unlike blogs or grey literature, these sources provide verifiable and credible data. The decision to include conference papers is informed by the common practice within the engineering disciplines, where peer-reviewed presentations are a primary method for disseminating cutting-edge research (Nicholas et al. 2015; Phillips et al. 2023).

Empirical Research in Higher Education Contexts (IC3): This study focuses exclusively on empirical research involving participants at higher educational levels – specifically undergraduate, master's, and doctoral students. The research examines the challenges of implementing design thinking within specific educational settings such as courses and projects.

Consideration of Design Thinking (IC4): Given the broad scope and diverse definitions of design thinking, this criterion mandates the inclusion of studies that describe and evaluate the implementation of design thinking in higher engineering education. To be included, research must explicitly mention design thinking in the title, abstract, or keywords, indicating that it is a focal point of the study. This requirement is intended to ensure that only reliable and relevant studies are analysed.

At Least One Engineering Discipline (IC5): Documents considered for this review must be centrally focused on engineering or STEM education, explicitly addressing at least one engineering or STEM discipline. The focus is defined to include: (i) general STEM fields, encompassing a broad range of disciplines such as mathematics, technology, and natural sciences; (ii) engineering or science categories; and (iii) specific STEM subjects.

These five inclusion criteria were developed to support us to answer the research questions. We also defined and adopted four types of exclusion criteria (EC):

Table 1. Databases and search strings used to locate articles.

Scopus: TITLE-ABS-KEY ("engineering education" OR stem* OR STEM*OR engineering* OR engineer* OR engineers* OR student* OR candidate OR candidates) AND ("design thinking") AND (learning OR education* OR university* OR curriculum OR pedagogy* OR approach) AND NOT ("elementary" OR "middle" OR "high" OR "k-12") AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "ch")) AND (LIMIT-TO (LANGUAGE, "English"))
Web of Science: TOPIC: ("engineering education" OR stem* OR STEM*OR engineering* OR engineer* OR engineers* OR student* OR candidate OR candidates) AND TOPiC: ("design thinking") AND TOPIC: (learning OR education* OR university* OR curriculum OR pedagogy* OR approach) AND TOPIC: (student OR students OR candidate OR candidates) NOT ("elementary" OR "middle" OR "high" OR "k-12")
ERIC: ("engineering education" OR stem* OR STEM*OR engineering* OR engineer* OR engineers* OR student* OR candidate OR candidates) AND ("design thinking") AND (learning OR education* OR university* OR curriculum OR pedagogy* OR approach) AND NOT ("elementary" OR "middle" OR "high" OR "k-12")

Target Not Primarily Engineering (EC1): Studies that primarily focus on non-engineering or non-STEM fields, such as medicine or agriculture, are excluded. These fields have distinct educational frameworks and methodologies that differ significantly from those in engineering and technology-focused disciplines.

Theoretical Only (EC2): Studies that only discuss theoretical aspects or conceptual frameworks without empirical research or practical application are excluded. This ensures the inclusion of studies that offer actionable insights into the implementation of design thinking.

Limited Part of Design Thinking (EC3): Studies where design thinking is mentioned only peripherally – briefly referenced in passing, listed among many unrelated topics, or lacking substantive discussion of its concepts or practices – are excluded. This ensures focus on research where design thinking is a central theme rather than incidental reference. However, this criterion does not exclude in-depth studies of individual design thinking components (empathy, prototyping, collaboration, etc.) when these are explicitly framed within design thinking structures and contribute meaningful insights to design thinking implementation or theory, as design thinking is inherently multifaceted encompassing processes, mindsets, and competencies.

Without Clear Definitions of Design Thinking (EC4): Studies lacking a clear definition of design thinking will be excluded. Variations in the definition, however, are admissible, provided they are explicitly articulated and rigorously justified within the framework of the study.

3.1.3 Process of selecting

Using the search strings provided in Table 1, a total of 899 papers were initially identified across multiple databases: Web of Science (375), SCOPUS (108), and ERIC (78). After consolidating the data from these databases and using EndNote reference management software, 71 duplicate records were removed, resulting in a collection of 742 unique studies. The screening process was conducted in two stages by multiple independent reviewers to ensure reliability and minimise bias. The first two authors independently screened all 742 records against the inclusion and exclusion criteria. Disagreements were discussed and resolved through consensus. Records that could not be definitively excluded based on title and abstract alone were advanced to Stage 2.

The same two authors independently reviewed full-text articles against the complete set of inclusion and exclusion criteria. For each potentially includable study, reviewers documented: (1) whether it met all inclusion criteria, (2) if excluded, which specific criterion it failed, and (3) rationale for borderline cases. When disagreements arose, the third author served as an independent arbitrator, reviewing the disputed article and the two reviewers' assessments before making a final determination.

The screening process began by filtering articles based on the relevance of their titles and keywords. Subsequently, abstracts of all retrieved studies were examined to assess compliance with the inclusion criteria (IC). Keywords, abstracts, and introductions of the papers were analysed to determine their alignment with the IC. For IC3-5, where abstracts and introductions provided insufficient information, further analysis of the papers' other sections was conducted.

3.1.3.1 Screening of keywords, titles, and abstracts. The keywords, titles, and abstracts of 671 publications were screened. A total of 352 records were excluded based on predefined inclusion criteria. Exclusions were due to a lack of peer review as specified by Types of Publications (IC2), leading to 53 exclusions; irrelevance to Empirical Research in Higher Education Contexts (IC3), accounting for 176 exclusions; lack of relevance to Design Thinking (IC4), also resulting in 176 exclusions; and the failure to focus on at least one engineering discipline (IC5), with 61 exclusions.

3.1.3.2 Full-text sifting and review. Following a procedure similar to the abstract screening, full-text articles were downloaded and recorded in a Microsoft Excel spreadsheet for comprehensive evaluation. Of the 244 full-text articles assessed, 177 were excluded due to irrelevance to Empirical Research in Higher Education Contexts (IC3), which accounted for 45 exclusions; lack of relevance

to Design Thinking (IC4), resulting in 31 exclusions; and failure to focus on at least one engineering discipline (IC5), with 5 exclusions. Additional exclusions included articles where the target was not primarily engineering (EC1), leading to 2 exclusions; those only discussing theoretical aspects (EC2), resulting in 18 exclusions; and those providing a limited or peripheral perspective on Design Thinking (EC3), 49 exclusions. The final review for a clear definition (EC4) resulted in an additional 6 articles excluded.

3.1.3.3 Additional sampling. Similar to the methodologies employed by Gough, Thomas, and Oliver (2012), we examined the reference lists of the 68 selected articles to identify further studies that met our inclusion criteria but were potentially missed due to database search limitations or indexing issues (Lecy and Beatty 2012; Papaioannou, Sutton, and Booth 2016). This process led to the identification of an additional 8 articles that were subsequently included in the review. Each article underwent the same rigorous assessment process as the original set. The final compilation of analysed papers totalled 76, with 68 originating from the initial database retrieval and 8 from the subsequent reference list search, which is based on the PRISMA model (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) (Kitchenham 2004; Moher et al. 2009). The overall selection process is visually summarised in Figure 1.

3.2. Data extraction and data analysis

Upon finalising the inclusion of studies, we began by recording the descriptive characteristics of the studies, including basic identifying information. The first author independently coded each article following the coding classifications and dimensions listed in Table 2. This coded information was extracted verbatim from the manuscripts and analysed descriptively and presented in tables and graphs.

To ensure a comprehensive investigation, the research team developed a coding protocol guided by the curriculum characteristics framework outlined by van den Akker, Kuiper, and Hameyer (2003) and employed the Engineering Education Research (EER) classification method proposed by Finelli, Borrego, and Rasoulifar (2015). A qualitative synthesis, utilising Braun and Clarke (2006)'s thematic process, was applied to all 76 studies to explore the conceptual foundations of design thinking (what), focused on identifying the predominant models and definitions employed across studies. The analysis also investigated the rationale for incorporating design thinking into engineering curricula (why) for different stakeholders, including students, educators, and society. This process involved thematic clustering, the creation of new categories, and inductive analysis of free-text comments to identify recurring themes. The final data were generated in alignment with the study's objectives and goals (Ritchie et al. 2003), ensuring a systematic classification of all research until a comprehensive categorisation was achieved. Additionally, guided by Krippendorff (2018) and Vaismoradi and Snelgrove (2019), a qualitative content analysis method was adopted to address the implementation strategies (how). The detailed coding frameworks and corresponding schemas are documented in Table 3. To systematically synthesise the diverse literature on design thinking

Table 2. Descriptive coding framework and corresponding schema.

Coding Type	Coding Category	Details
Descriptive Coding	Basic Identification Information	AuthorT he year of publicationP ublication venues
	Research Methods	Qualitative Methods Quantitative Methods Mixed Method
	Practical Methods	Observational Experimental: Comparative or Validation

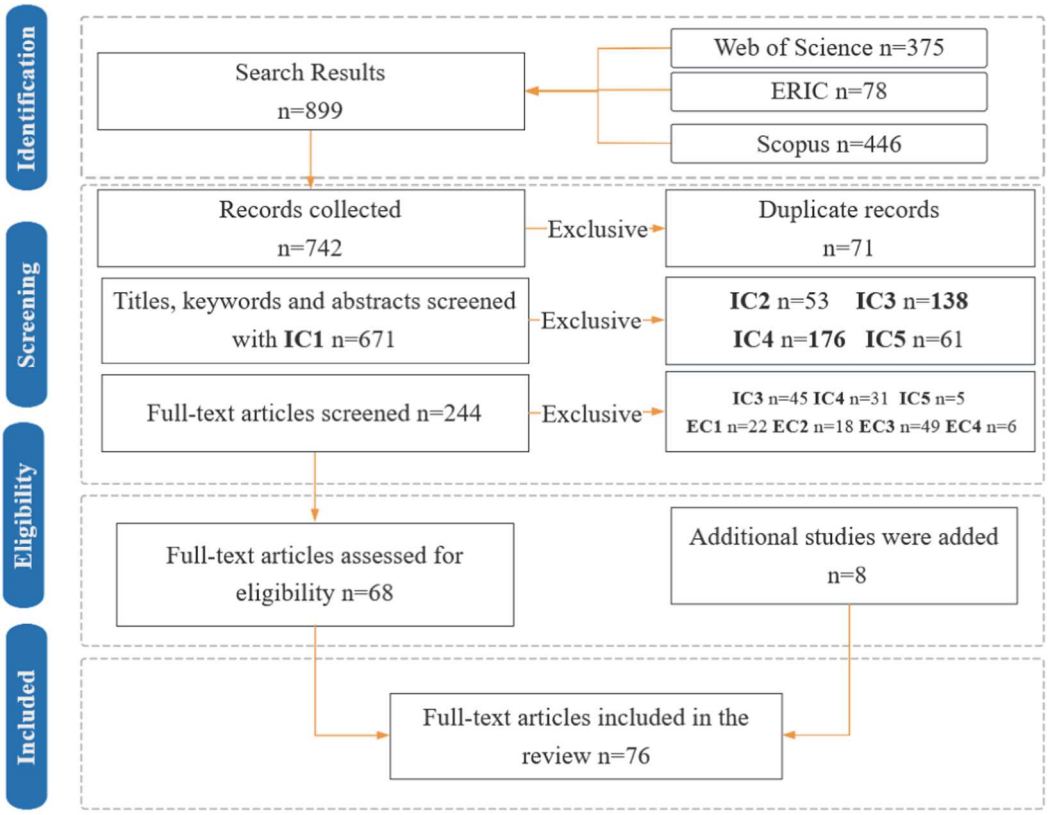

Figure 1. PRISMA flow chart highlighting the systematic literature review process (modified from Moher et al. (2009)) 22. Figure 1. PRISMA flow chart highlighting the systematic literature review process (modified from Moher et al. (2009)) 67.

in engineering education, we developed a comprehensive HOW-WHAT-WHY analytical framework. This multidimensional approach enables examination of design thinking simultaneously as a pedagogical practice (HOW it is taught), a conceptual construct (WHAT it comprises), and a purposeful intervention (WHY it is adopted). This framework draws on established traditions in implementation science (Nilsen 2020) and educational programme evaluation (Stufflebeam 2003) while being specifically adapted to the unique characteristics of design thinking pedagogy.

Table 3. Coding framework and corresponding schema.

Coding Category	Details		
The overarching foci of included research	Implementation Introduction of Design Thinking Design Thinking Effectiveness and VerificationDevelopment/ Enhancement of Design Thinking		
How	Implementation Characteristics	Learning Activities Collaboration Evaluation	
What	Most Popular DT Definitions Most Popular DT Models	Cited Sources	
Why	Rationale for Implementation	For Students	Knowledge Acquisition Emotional Needs
		For Educators	Educator Training & Support Teaching Tool Development
		For Society	Sustainability Development Goals Social Justice

Addressing concerns about qualitative research subjectivity raised by some researchers (Mayring 2014; Schreier et al. 2020), we analysed coding frequencies (Borrego, Foster, and Froyd 2014) to enhance objectivity and transparency in our synthesis. It is critical to note that the research themes delineated were not strictly exclusive; findings from individual studies often spanned multiple thematic areas. To address this, the team employed a stringent analysis methodology, ensuring precise classification of each study into the most relevant research themes (Ritchie et al. 2003). This process facilitated a robust interpretation of the data, anchored by the thematic statements.

3.2.1 Bias control

Multiple strategies were implemented to ensure reliability and minimise bias throughout the study selection, data extraction, and analysis processes.

To ensure the reliability of study selection, the first and second authors conducted independent eligibility assessments at both major screening stages (title/abstract screening and full-text review), following established protocols for systematic reviews in engineering education research (Borrego, Foster, and Froyd 2014). Inter-rater reliability was assessed using Cohen's Kappa (κ) statistic (Cohen 1960), which measures agreement between two raters while accounting for agreement that might occur by chance. Cohen's Kappa was calculated for both screening stages:

Title/Abstract Screening Stage: $\kappa = 0.82$ (95% CI: 0.78-0.86), indicating substantial agreement according to conventional interpretation guidelines (Landis and Koch 1977). Full-Text Review Stage: $\kappa = 0.85$ (95% CI: 0.79-0.91), indicating almost perfect agreement (Fleiss, Levin, and Paik 2013; Landis and Koch 1977). These high agreement levels demonstrate robust inter-rater reliability and suggest that the inclusion and exclusion criteria were clearly defined and consistently applied across reviewers. Of the 244 full-text articles reviewed in Stage 2, initial disagreements between the two primary reviewers occurred for only 18 articles (7.4% disagreement rate). All disagreements were successfully resolved through either structured discussion leading to consensus ($n = 12$) or through independent arbitration by the third author ($n = 6$). No disagreements remained unresolved, and no articles required exclusion due to inability to reach consensus.

3.3. Validity, reliability, and limitations

Despite the large number of studies reviewed, several measures were implemented to ensure the validity of the review process. Following Borrego, Foster, and Froyd (2014), a comprehensive search strategy was employed to minimise the risk of missing relevant literature by using a combination of keywords. And snowball sampling, as suggested by Gough, Thomas, and Oliver (2012), was used during the full-text review stage to examine the reference lists of the selected articles, further reducing the likelihood of omissions. In cases where abstracts and introductions lacked sufficient information to determine relevance, additional sections of the papers were analysed to ensure a thorough evaluation. During the article selection process, three authors independently reviewed the titles and abstracts of the retrieved papers, applying the predefined inclusion and exclusion criteria. When discrepancies arose regarding the inclusion of specific articles, resolution was sought through structured discussions among the authors. These meetings served not only to resolve disagreements but also to refine the selection criteria as needed, thus enhancing the consistency of the review process.

We acknowledge several limitations in the methods used in this study. Although design thinking is inherently interdisciplinary, our inclusion criteria specifically focused on engineering disciplines to maintain relevance to our research question regarding the implementation of design thinking in higher engineering education. This focus may have excluded valuable insights from studies that span multiple STEM disciplines or that apply design thinking in a broader context beyond engineering. Consequently, the strict focus on engineering may have narrowed the scope of our review and potentially left out relevant interdisciplinary research that could have contributed to a more comprehensive understanding.

4. Results

Initially, we outline the descriptive characteristics including publication timelines, venues, study types, methodologies, and target disciplines. We then present the qualitative synthesis of research aims and theoretical definitions of design thinking. This is followed by an examination of design thinking implementation at both course and project levels. Finally, we highlight insights into the core principles and objectives of design thinking. Detailed descriptive characteristics and coding information are available in the Supplementary Tables in the accompanying Excel.

4.1. Descriptive characteristics

4.1.1 Publication timeline

Figure 2 illustrates that the reviewed sample includes 76 works, comprising 44 journal articles and 32 conference proceedings. Spanning a 20-year review period, the earliest work appeared in 2006 (McAlpine, Reidsema, and Allen 2006), following Dym et al.'s (2005) introduction of the concept of engineering design thinking. During the formative phase (2005–2013), research activity was limited to just 7 publications, primarily consisting of journal articles focused on theoretical frameworks and pedagogical models. The transitional phase (2014–2019) marked a period of accelerated growth, driven by conference proceedings, which dominated this phase and reflected the emphasis on emerging trends and methodological experimentation. In the pandemic-driven acceleration phase (2020–2023), publication output surged, accounting for nearly half of the total publications. During this period, journal articles regained dominance, reflecting a shift toward more in-depth and refined investigations. The COVID-19 pandemic catalysed research innovations, with the rapid shift to online and hybrid learning environments leading to studies on virtual collaboration, remote prototyping, and the use of digital design tools. Both journal articles and conference proceedings display a consistent upward trajectory, highlighting the growing recognition of design thinking within engineering curricula and academic research. The sustained momentum suggests that research in this field will continue expanding beyond 2023, with an increasing emphasis on multidisciplinary applications, innovative methodologies, and real-world impact.

4.1.2 Publication venues

Among the 76 papers reviewed, 44 are journal articles and 32 are conference papers. These publications span diverse academic fields, reflecting design thinking's interdisciplinary nature in higher engineering education contexts. To understand the disciplinary homes of design thinking research, we categorised publications into five primary fields: Education, Engineering, Design, Management, and Education Technologies. Table 4 presents the distribution of publications across these fields, organised separately for journal articles and conference papers. This distribution pattern suggests that design thinking in higher engineering education has established a clear disciplinary home in

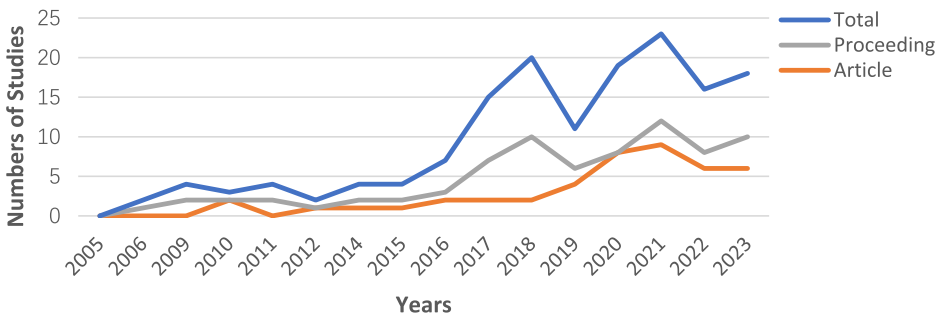

Figure 2. Temporal distribution of design thinking publications in higher engineering education (2003–2023) 26.

engineering education literature while maintaining productive connections to adjacent fields including general education, design studies, education technologies, and engineering disciplines. The predominance of engineering education venues provides coherent knowledge base for the field while cross-disciplinary publications facilitate broader conversations about design thinking's role in higher education.

4.1.3 Research methodologies and study designs

Table 5 categorises the reviewed studies by research methodology, revealing important patterns about how design thinking in engineering education is investigated.

4.1.3.1 Type of study design. In this review of 76 studies, 53 are observational, primarily focusing on describing teaching processes within design thinking. These studies typically observe subjects in natural settings or specific scenarios to gather insights into the application and outcomes of design thinking. There are also 23 experimental studies, which are further divided into 14 comparative experiments and 9 validation experiments. These studies explore the efficacy and applicability of various design thinking approaches. 8 longitudinal studies assessed the impact of design thinking interventions on student achievement or attitudes. All other studies were either cross-sectional or pretest–posttest studies, including 3 articles analysed design thinking pedagogy in comparison to traditional or innovative educational methods (Charosky et al. 2022; Corral and Fronza 2018; Liu, Lee, and Huang 2023). And Todoroff et al. (2021) compared design thinking among students in different STEM majors (civil engineering and architecture). Additionally, Suligoj, Zavbi, and Avsec (2020) explored gender differences and Coleman et al. (2020) assessed variations across different grade levels.

4.1.3.2 Method of practical studies. In terms of research methods, 40% of the studies (30 of 76) were classified as descriptive analyses of design thinking, presenting specific courses or environments. These studies detail the design processes and outcomes of design thinking courses, with particular emphasis on deliverables. Qualitative research methods were exclusively or predominantly used in 30% of the studies (26 of 76). Following Bork and Mondisa (2022) categorisation, another 15% of the studies (11 of 76) employed exclusively or predominantly quantitative research methods. These quantitative studies utilised various statistical analyses, including descriptive statistics such as mean and median (Isa et al. 2019; Rak 2023; Soomro, Casakin, and Georgiev 2021; Srisawat and Wannapiroon 2022), inferential statistics (Chen, Cheng, and Wu 2023; Isa et al. 2019) like t-tests and regression analyses, hypothesis tests for between-groups interventions (Coleman et al. 2020; Guaman-Quintanilla

Table 4. Distribution of publications across academic fields and venues.

	Category	Sub-category	*N*		Category	Sub-category	*N*
Journal Papers	Education Fields	Engineering Education	21	**Conference Papers**	Education Fields	Engineering Education Conferences	16
		Science and Technology Education	1			Education Technologies	4
		STEM Education	2			Design Education	1
		General Education	3			General Education	1
		Education Technologies	6		Engineering Fields	Software/ Information Systems/ Mechanical Engineering/ Mechatronics	5
	Engineering Fields	Sustainability/Software/ Mechanical Design/ Biomechanical	6		Design Fields	Design and Product Development	5
	Design and Creativity Fields		4		Total		32
	Management Fields		1				
Total			44				

Table 5. Categorisation of the research by methodologies.

Type of Study Design	N = 76	Method of Practical Studies	N = 76
		Descriptive	30
Observational	53	Qualitative	26
		Quantitative	11
		Mixed	9
Experimental:	23	**Target Disciplines**	**N = 76**
Comparison	14	Engineering	59
		Science	3
Validation	9	Science and Engineering	5
		STEM	9

et al. 2023), and modelling such as factor analysis (Arastoopour Irgens et al. 2015). The remaining 9 papers reported use of both quantitative and qualitative research methods.

4.1.3.3 Target disciplines. The target disciplines in this review are categorised into four distinct groups: (i) only about engineering, engineering/specific disciplines within engineering (59), such as bioengineering, (Chen, Cheng, and Wu 2023), electrical engineering (McKilligan et al. 2017), software engineering (Palacin-Silva et al. 2017), civil engineering (Isa et al. 2019), mechanical engineering (Lande and Leifer 2010) and design engineering (Ericson et al. 2009a); (ii) only about science, science /specific disciplines in science that focus on physics (Luna, Talavera, and Chong 2018; Lur, Cheah, and Lee 2022) and programming (Lai et al. 2022); (iii) both science and engineering; (iv) about STEM fields as an aggregate (Lin et al. 2021; Malele and Ramaboka 2020).

From Figure 3, it is clear that the sequence of research disciplines evolved from Engineering (E) to Science and Engineering (S + E), then to a broader development in STEM disciplines, and finally focusing on Science (S). This progression illustrates how design thinking initially took root within engineering and subsequently extended its influence across other STEM disciplines (Purzer and Quintana-Cifuentes 2019). After 2017, a distinct trend towards integration became apparent, with increasing numbers of publications in combined Science and Engineering categories, as well as broader STEM fields. Publications related to these broader STEM categories started appearing in 2014 but were initially sporadic. However, from 2020 onwards, there was a significant increase in publications within these categories, indicating a substantial shift towards a more integrated approach in STEM research.

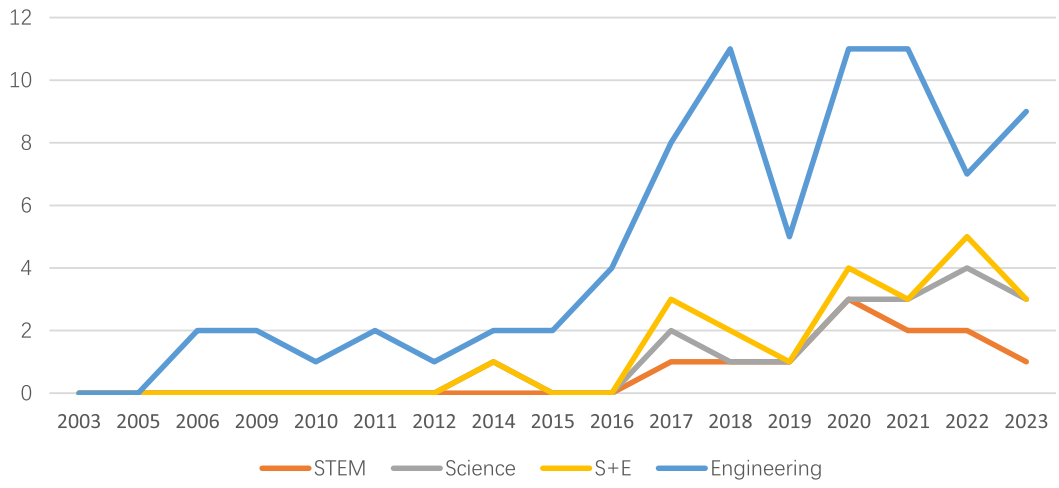

Figure 3. Studies Published of each discipline between 2003 and 2023 68.

4.2. Qualitative synthesis of findings

Using thematic analysis techniques, we identified three overarching themes related to design thinking in higher engineering education from the findings of the included studies. Included (i) Implementation Introduction of Design Thinking (33/76): This theme focuses on how design thinking is integrated into education. It examines various methods and strategies for incorporating design thinking principles into educational curricula and activities. Understanding different methods of implementation can shed light on best practices for educators. (ii) Design Thinking Effectiveness and Verification (34/76): This theme explores the transformative impact of design thinking on learning outcomes, student engagement and problem-solving capabilities, particularly within engineering disciplines. (iii) Development/Enhancement of Design Thinking (9/76): This theme centres on the technological and methodological advancements that facilitate the development of design thinking skills. It encompasses the development of e-learning platforms (Ostrowski et al. 2015) alongside testing standards (Shah, Smith, and Woodward 2009), software solutions (Arastoopour Irgens et al. 2015), online platform (Srisawat and Wannapiroon 2022), and tools designed to support the effective teaching and application of design thinking principles (Souza et al. 2020).

To provide a clearer explanation, Figure 4 illustrates these key trends through a three-dimensional HOW-WHAT-WHY framework. The subsequent sections will examine each dimension in detail, covering the critical aspects reflected in the visual representation.

4.2.1 WHAT most-cited sources of design thinking

4.2.1.1 Most-cited definitions of design thinking. The term design thinking itself can be understood and interpreted in quite different ways (Goldschmidt and Rodgers 2013; Kimbell 2011), and even within the field of education alone the understanding changes with different users or application scenarios (Micheli et al. 2019). Although the criteria we reviewed were the necessity to provide a clear or explicit definition of design thinking, we found that 5 did not cite any design thinking sources to define design thinking. As shown in the Table 6, in the remaining 67 papers, three definitions of design thinking were cited most frequently, each highlighting a unique focus that is particularly important in higher engineering education. Based on the frequency of citations and content analysis, these definitions can be categorised into three main areas of focus.

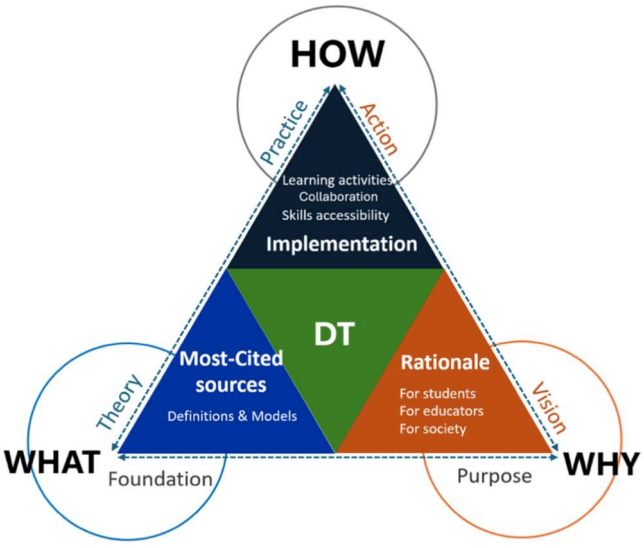

Figure 4. Three-dimensional model of design thinking education: HOW-WHAT-WHY framework 68.

The literature predominantly references three conceptualisations of design thinking. Tim Brown's definitions (Brown 2008; Brown and Katz 2009), cited in 60% of studies, frame it as a human-centred innovation process integrating user needs, technological feasibility, and business viability. This perspective highlights cognitive strategies like empathy, user-centred problem framing, and balancing practical constraints with creative solutions. The second prominent definition, proposed by Dym et al. (2005) and cited in 30% of works, emphasises practical competencies for engineering design education. Core skills include managing ambiguity, convergent-divergent thinking, systems thinking, teamwork, and effective communication of design concepts. A third cluster (20%) employs Buchanan (1992)'s framework to position design thinking as an experimental counterpoint to traditional methodologies. It distinguishes design thinking from traditional methodologies in the natural and social sciences by emphasising its experimental essence. Kimbell (2011) underscores its role in challenging existing conditions, generating innovative possibilities, and addressing wicked problems through active experimentation. This approach promotes creativity and the practical construction of models and solutions, affirming the role of design thinking as an indispensable tool for solving complex problems in a dynamic world.

The coexistence of these three conceptualisations within the engineering education literature reflects both richness and tension. This diversity represents a form of theoretical pluralism in which different definitions serve distinct pedagogical aims. Brown's process-oriented framework aligns with innovation-driven curricula, Dym et al.'s competency-based model informs structured curriculum design, and Buchanan's epistemological framing provides philosophical grounding for educational reform. Yet this plurality also generates ambiguity regarding what design thinking entails and how it should be taught. The variation in how these frameworks are cited suggests that engineering educators tend to draw selectively from available definitions according to their pedagogical contexts and goals, rather than converging toward a shared conceptual understanding.

4.2.1.2 Most-applied models of design thinking. Although design thinking lacks a universally standardised methodology, the Stanford Design Thinking Model (d.school) (Camacho 2016), is the most widely adopted framework in educational research. This model structures problem-solving into five iterative phases: empathy (understanding user needs), problem definition (reframing challenges), ideation (generating creative solutions), prototyping (building tangible representations), and testing (evaluating and refining outcomes). Its emphasis on cyclical iteration and user-centred focus positions it as a foundational tool for innovation (Goudarz Poursharif et al. 2021). Some studies adapt the model to specific requirements rather than adopting it directly. For example, some studies condense the process into core stages (e.g. empathy, ideation, prototyping) to enhance pedagogical efficiency in time-constrained settings (Palacin-Silva et al. 2017). Other studies extend the traditional five-step model by adding detailed steps such as implementation (Yakub, Naik, and Unnisa 2018) or further refinement (Kaygan 2023; Ranger and Mantzavinou 2018). In addition, the design thinking process is often combined with other approaches, such as the Conceive-Design-Implement-Operate framework (CDIO) (Charosky et al. 2018; Isa et al. 2019; Lai et al. 2022; Ping, Chow, and Teoh 2011; Ranger and Mantzavinou 2018), problem-based learning (PBL) (Colmenares-Quintero et al. 2023; Isa et al. 2019; Kuo et al. 2021; Goudarz Poursharif et al. 2021) or the Two Diamonds model (Guaman-Quintanilla et al. 2023), to create a more holistic approach to design thinking that meets specific educational needs and contexts.

Table 6. Most-Cited definitions of design thinking.

Author	Year	Frequency of Citation in the Sample
Brown	2008, 2009	58%
Dym	2005	33%
Buchanan	1992	18%

4.3. HOW: implementation characteristics

4.3.1 Learning activities

Design thinking practices are implemented across three primary levels in higher education: course-level integration (31/76), project applications (22/76), and instructional strategy frameworks (23/76). At the course level, design thinking is embedded in both semester-long modules and short-term intensive programmes. Semester-long courses, such as the 27 undergraduate programmes reviewed, emphasise iterative problem-solving through cycles of prototyping and testing, often supported by technical tools like CAD software or simulations. These courses are tailored to students' academic progression, with 11 programmes explicitly designed for first-year undergraduates to introduce foundational principles, while others target advanced cohorts (5 for second-year, 2 for third-year students) or graduate and doctoral learners, adapting content to prior experience and career goals. Complementing these long-term offerings are short-term courses (less than a week), which focus on rapid skill acquisition through condensed lectures and hands-on activities, such as developing brainstorming techniques or addressing narrowly defined problems. Typically offered as electives outside core curricula, these courses prioritise flexibility, enabling students to engage based on personal interest and fostering self-directed learning.

Beyond formal courses, design thinking is operationalised through experiential projects. Among the 76 studies reviewed, 22 highlight project-level implementations, including capstone projects and short-term challenges that immerse students in competitive, time-bound problem-solving scenarios (e.g. hackathons). These initiatives cultivate critical analysis and teamwork, often serving as students' first exposure to open-ended, real-world problems. Additionally, innovation ecosystems – comprising physical makerspaces, digital platforms, and interdisciplinary labs – provide environments for prolonged collaboration. These spaces equip students with prototyping tools and resources, bridging academic departments and industry partners to tackle complex challenges through iterative experimentation.

A third dimension involves integrating design thinking as a systemic pedagogical framework, evident in 23 studies. Here, the approach transcends isolated courses or projects, becoming a mindset woven into curriculum design. Open-ended, self-directed projects encourage students to apply design principles to personal or societal interests, such as entrepreneurship or sustainability, fostering creative autonomy. Research-driven strategies further translate theoretical insights into practice, exemplified by the integration of design thinking with established educational frameworks like Problem-Based Learning (PBL) or the Conceive-Design-Implement-Operate (CDIO) model. These hybrid approaches aim to refine teaching methodologies – for instance, redefining assessment criteria for iterative tasks or embedding reflective practices – to align classroom activities with real-world innovation demands. By merging technical skill development (e.g. prototyping, systems thinking) with soft skills (e.g. empathy, communication), this tripartite model ensures design thinking's adaptability across disciplines and academic stages, positioning it as both a pedagogical tool and a educational philosophy in addressing dynamic global challenges.

4.3.2 Teamwork and collaboration

Design thinking is typically taught within a team-work or collaborative environment requiring students to leverage their diverse perspectives and expertise (de Carvalho et al. 2020). Collaboration is realised in distinct ways: Intra-disciplinary, inter-disciplinary and external collaboration with staff or industry.

In multidisciplinary team projects, students with similar academic backgrounds work together, and peer learning is enhanced by supervision and critique (Dotson et al. 2020; Levine, Lesniewski, and Agogino 2016). Collaboration among student teams from various STEM disciplines, such as combining expertise in computer science, electrical engineering, and mechanical engineering, increases the appeal and feasibility of product development requirements (Ibwe et al. 2018; Kuo et al. 2021). Malele and Ramaboka (2020) emphasised the importance of students from various areas playing

specialised roles in interdisciplinary design studio courses. These roles promote cross-disciplinary collaboration and improve the overall learning experience.

Collaboration with non-STEM students: integrating humanistic perspectives through collaborations with non-STEM disciplines (e.g. business, or liberal arts) is critical to enriching the design process (Kaygan 2023). Such interdisciplinary collaborations are essential as they often lead to more original and creative solutions by blending diverse perspectives and expertise. Crawley et al. (2007) emphasised that engaging with humanities and business schools allows STEM students to appreciate and apply the societal and commercial implications of their projects. Through collaboration experience, students can understand innovation as a methodical design process with unique tools and techniques (Ranger and Mantzavinou 2018). Kaygan (2023) and Levy (2017) highlighted that cross-disciplinary cooperation extends beyond technical solutions by embracing contributions from sociology, business, design, and the arts. This broadens the educational scope and enhances the generation of novel ideas. Such interactions enrich the learning experience, making it more comprehensive and relevant to real-world contexts.

Faculty team collaboration: Certain studies suggest the inclusion of educators with design expertise in teaching design thinking courses. For example, in engineering courses (Fila, McKIlligan, and Guerin 2018; McDonald et al. 2019; McKilligan et al. 2017), educators from engineering, science, and design collaborated to support the curriculum design approach for engineering education with design thinking principles. However, it is important to acknowledge that collaborative teaching across disciplines is still uncommon in higher education due to traditional departmental structures and communication challenges (Bairaktarova and Plumlee 2022).

Engineering industry professionals' collaboration: Engaging industry professionals significantly enhances the relevance and efficacy of continuing education programmes in engineering fields (Luna, Talavera, and Chong 2018). Incorporating external stakeholders from various industries introduces students to innovative business contexts and helps them understand the interplay between technical prowess and commercial viability. The adoption of design thinking pedagogies empowers students with the entrepreneurial skills needed to commercialise technologies. Charosky et al. (2022) emphasised company involvement, providing practical experiences enhancing responsibility and engagement. According to Dotson (2020), such collaborations with local businesses not only extend the reach and impact of educational programmes but also ensure their sustainability and scalability.

4.3.3 Design thinking skills and assessment

Design thinking, as both a cognitive paradigm and cultural practice, is rooted in distinct psychological frameworks that prioritise human-centred problem-solving (Brenner, Uebernickel, and Abrell 2016). Contemporary research systematically examines its cognitive, affective, and behavioural dimensions, underscoring the critical need for early cultivation of associated competencies (Bairaktarova and Plumlee 2022; Jiang and Pang 2023). This analysis synthesises findings from 70 empirical studies (excluding six non-assessment focused works) to establish an evidence-based framework for evaluating four core design thinking competencies: empathy, collaboration, creativity, and problem-solving. The corpus reveals a methodological pluralism in assessing design thinking skills, as demonstrated in Table 7. Although the total number of assessments is 70, the combined percentages exceed 100% due to the overlapping use of multiple methods within the same studies. Many studies integrate several approaches, such as pairing surveys with interviews or combining design projects with reflective journals. This overlap reflects the complex nature of design thinking, where no single method can fully capture the range of cognitive, affective, and practical skills involved. The integration of multiple assessments enables both formative and summative evaluations, ensuring a comprehensive understanding of students' problem-solving growth and overall development.

Several methodological patterns can be identified across the reviewed studies. Structured quantitative approaches such as surveys (41%) and rubric-based design projects (36%) are the most used.

These methods provide standardisation and scalability, allowing researchers to assess large student cohorts efficiently and measure specific competencies in a consistent manner. Complementing these, qualitative approaches including interviews (29%) and analyses of reflective documents (20%) offer richer insights into students' engagement with design thinking. Such approaches capture nuances often missed by quantitative tools, such as how learners define problems, navigate ambiguity, and develop empathy through iterative engagement. Despite these advances, direct observation remains rarely employed, appearing in only 4 percent of studies. This gap is noteworthy, as observational and video-based analyses provide valuable opportunities to capture real-time collaboration, problem-solving behaviours, and other tacit dimensions of design thinking that participants may not articulate through self-reports. The limited use of observation likely reflects practical constraints, including time-intensive data collection, the need for trained analysts, and typically smaller sample sizes, which can hinder generalisability. However, advances in video analysis technologies and systematic coding frameworks offer promising avenues for incorporating behavioural evidence into design thinking research. Integrating such data with existing survey and interview findings could enhance methodological triangulation and deepen understanding of design thinking as a dynamic, situated learning process.

4.3.3.1 Empathy. Empathy and a user-centred approach are key components of design thinking, which focuses on these ideas to understand the needs, wants and experiences of users (Ping, Chow, and Teoh 2011). Empathy is defined as a student's ability to observe, participate, watch, and listen to understand the needs and preferences of people and society, enabling them to take informed and targeted actions (Cooper 2011). By developing empathy for users and immersing themselves in their perspectives, engineering students can gain a deep understanding of real-world problems (Bairaktarova and Plumlee 2022) and improve their perceptual skills (Todoroff et al. 2021). Empathising with stakeholders is the first phase in the design thinking process and is a prerequisite for user-centred design (Lucas, Claxton, and Hanson 2014; Shetty and Xu 2018). Previous studies have demonstrated that higher engineering education, when approached empathetically, enables students to better understand user needs and develop more effective solutions (Luna, Talavera, and Chong 2018; Rasoal, Danielsson, and Jungert 2012). Role-playing and experiential learning were shown to help students develop empathy for stakeholder groups, shifting their focus from solely technical solutions to enhancing user comfort, safety, and well-being (Lechuga, Clerc, and Howell 2009). For instance, Yakub, Naik, and Unnisa (2018) developed a project based on a design thinking framework, crafting a weight-measurement scale specifically designed for elderly users, which considers both apparent and potential user issues. By cultivating empathy, students can fully grasp the user-centred concept, understanding contexts, goals, and challenges. Galleguillos, Santelices, and Bustos (2019) and Al-Qaralleh, Sababha, and Abugharbieh (2021) emphasised the importance of a human-centred approach in designing solutions that are technically sound and aligned with user needs.

Assessing empathy in engineering design education remains a complex challenge due to its affective and behavioural dimensions. Immersive activities allow students to capture subtle insights

Table 7. Distribution of assessment methodologies across reviewed studies.

Category	Method	Evaluation Tool	Count	Percentage (%)
Structured Quantitative Evaluation Tools	Survey (questionnaires)	Likert-scale questionnaires	29	41%
	Design project (essay, prototypes, course artefacts)	Rubrics or rating scales (expert evaluation matrix)	25	30%
Contextualised Qualitative Assessment Methods	Interview (open-ended questions, semi-structured interviews)	Thematic analysis, content analysis	20	27%
	Document analysis (self-reflections)	Self-reflection analysis	14	18%
	Observation (in natural settings)	Video-based observation (supporting other methods)	3	6%

that traditional surveys often overlook. Colmenares-Quintero et al. (2023) and Palacin-Silva et al. (2017) argued the use of personas, user storyboards, and empathy maps as tools for understanding user needs. While Rasoal, Danielsson, and Jungert (2012) pioneered the adaptation of the Interpersonal Reactivity Index (IRI) for engineering contexts, subsequent studies reveal limitations in relying solely on self-reported perspective-taking and empathic concern scores. Advanced methodologies now integrate behavioural analytics through coded interview transcripts and artefact evaluation of empathy maps (Colmenares-Quintero et al. 2023). The Yakub, Naik, and Unnisa (2018) geriatric scale design project demonstrates how synthetic assessment frameworks can triangulate user-centred solution quality with process documentation, bridging the gap between stated empathic orientation and demonstrated empathic practice.

Collaboration

Collaboration consistently emerges as a central pedagogical mechanism in DT-based engineering education. Empirical evidence suggests that integrating design thinking into curricula enhances students' teamwork, communication, and co-creation abilities (Ericson et al. 2009b). The iterative and participatory nature of design thinking encourages students to explore roles within teams and to engage in shared reflection and feedback (Guerra and Shealy 2018; Mikic and Rudnitsky 2016; Somerville and Townsend 2014). As indicated by Tiwari and Rathore (2022) participation in design thinking projects appears to foster students' confidence in collaboration and collective problem-solving.

Across studies, design thinking's interdisciplinary teamwork model is highlighted as a catalyst for integrative learning. When engineering, management, and design students collaborate, they collectively negotiate diverse viewpoints and co-develop creative, socially responsible solutions (Charosky et al. 2022; Leem and Lee 2015; McDonald et al. 2019). Such teamwork mirrors professional engineering contexts, where innovation often depends on cross-domain communication and mutual adaptation (Zancul et al. 2017).

Project-based and CDIO frameworks are frequently cited as structural enablers of effective collaboration (Goudarz Poursharif et al. 2021). Increasingly, digital collaboration, through online platforms and virtual studios, has expanded opportunities for distributed teamwork and ongoing feedback (Liu, Lee, and Huang 2023; Rahman et al. 2019). Overall, collaboration functions not only as a pedagogical strategy but also as a measurable outcome of design thinking education, supporting the development of communication, empathy, and problem-solving capacities essential to twenty-first-century engineering practice.

4.3.3.2 Problem solving. The goal of design thinking training is to equip engineering students with a comprehensive problem-solving approach, including how to construct and address unconventional problems (Dzombak and Beckman 2020). Design thinking is characterised as a collaborative and cyclical process encompassing problem definition, ideation, prototyping, testing, and refinement (Panke 2019). This structure helps students approach ill-defined problems systematically, developing both creative and analytical thinking skills essential for engineering practice.

Assessment of problem-solving in DT-based education typically focuses on both the process and the quality of the outcomes. Design projects serve as key assessment mechanisms for evaluating students' ability to define problems, generate innovative ideas, and iterate through prototyping and testing (Lur, Cheah, and Lee 2022). Charosky and Bragos (2021) emphasised the role of design thinking in addressing complex societal challenges, aligning engineering education with real-world applications, and reinforcing its relevance and impact. G. Poursharif et al. (2021) integrated design thinking into engineering curriculum, equipping students with skills crucial for lifelong learning and the capacity to effect societal change. By solving open-ended design challenges and addressing critical social issues, students not only enhance their learning experience but also prepare to contribute meaningfully to various societal contexts (Guaman-Quintanilla et al. 2023; Lucas, Claxton, and Hanson 2014). Liu, Lee, and Huang (2023) and Guaman-Quintanilla et al. (2023) noted that engineering students gain the capacity to handle complex real-world situations, encourage creativity, accept

design challenges, participate in lifelong learning, and ultimately produce original solutions. Assessment typically evaluates students' ability to identify complex problems, explore multiple solutions, and develop iterative prototypes or models within a given timeframe.

4.3.3.3 Creativity. The centrality of creativity in higher engineering education is widely recognised, emphasising innovation, invention, and the capacity for creativity as core elements (Açar and Rother 2011; Kuo, Tseng, and Yang 2019; Shetty and Xu 2018; Ulibarri et al. 2014). Creativity in this context involves risk-taking, curiosity, inventiveness, and embracing challenges (Jiang and Pang 2023). Research by Luna, Talavera, and Chong (2018) suggest that design thinking catalyses creativity in problem-solving processes by encouraging students to deviate from conventional thinking patterns. Similarly, McElheron and Pilgaard Harsaae (2016) contend that this approach fosters the emergence of original and innovative ideas, enhancing the adaptability and creative capabilities of engineering students to tackle complex challenges. Jiang and Pang (2023) further emphasised that design thinking serves as an effective tool for fostering innovation capabilities and enhancing the application of advanced engineering solutions. Extensive research supported that design thinking not only increases creativity among engineering students but also enables them to generate pioneering ideas in their respective fields (Bhatnagar and Badke-Schaub 2017; Gaston, Guffey, and Rand 2023). Additionally, design thinking enhances students' self-awareness of their inventive capacities (Charosky et al. 2022; McAlpine, Reidsema, and Allen 2006). Empirical evidence suggested that the integration of design thinking into educational practices positively influences students' motivation and bolsters their capacity for innovation (Corral and Fronza 2018; Kuo et al. 2022; Lande and Leifer 2009; Mikic and Rudnitsky 2016).

In terms of assessment, surveys remain the most common method for evaluating creative confidence, originality, and flexibility. Tools such as the Competency-Based Creative Agency Scale (Ulibarri et al. 2014) and creativity tendency scales (Jiang and Pang 2023) provide quantitative measures, while qualitative methods such as reflective journals, interviews, and portfolio analyses allow deeper exploration of students' creative processes. Coding and thematic analysis of reflections and design artefacts can reveal how students iterate, adapt, and internalise creative thinking strategies (Clark, Stabryla, and Gilbertson 2020). Collectively, these findings suggest that creativity in DT-based engineering education is not only a product of individual talent, but a process nurtured through structured exploration, reflection, and collaboration.

4.4. WHY: evaluation of benefits of design thinking integration

This section considers the evidence for design thinking benefits at both individual (micro) and organisational (macro) levels. At the micro level, design thinking focuses on developing students' individual capabilities and competencies including creativity, empathy and problem solving. At the macro level, design thinking impacts three key stakeholder groups: students, educators (representing educational institutions), and society. Table 8 provides a structured analysis of design thinking impact across various sectors within higher engineering fields.

4.4.1 For students

4.4.1.1 Knowledge acquisition: disciplinary and interdisciplinary perspectives. Design thinking, via emphasis on collaboration, enhances the acquisition of in-depth disciplinary knowledge combined with broader interdisciplinary understanding. In engineering courses, it encourages exploratory approaches that elevate students' mastery of existing knowledge, overcoming traditional boundaries to establish new connections. This approach not only deepens their expertise in specific fields but also broadens their understanding of how different disciplines intersect and interact. Such interdisciplinary engagement enriches the educational experience, making it more comprehensive and integrative.

There are many benefits of multidisciplinary education, including the opportunity for students to gain insights and knowledge from peers and educators in different fields that they may not have encountered in a single discipline (Al-Qaralleh, Sababha, and Abugharbieh 2021). Thomas et al. (2021) suggested that cross-disciplinary courses that transcend traditional boundaries allow students from different fields to acquire knowledge applicable to addressing problems within their specialised areas. The limitations of single-subject learning include the potential reliance on stereotypical thinking and to restrict thinking and behaviour by relying solely on one area of expertise (Forest et al. 2014). For instance, Suligoj, Zavbi, and Avsec (2020) 's survey indicated that interdisciplinary methods could enhance the comprehension of students in electrical engineering and computer science majors. Some studies provide specific examples of interdisciplinary integration, highlighting how design thinking courses enable engineering students to engage with cultural elements of the liberal arts and embrace different perspectives. Malele and Ramaboka (2020) described how these courses encourage students to consider the liberal arts, thereby fostering a broader understanding of different problem-solving approaches. This exposure not only stimulates student interest but also brings new perspectives to the traditional rigid thinking prevalent among engineers. Furthermore, Henriksen, Mehta, and Mehta (2019) supported the notion that such educational frameworks can enrich the learning experience. The STEM-Arts integration model is a good example of how, by combining art and design with STEM disciplines, it fosters interdisciplinary research and collaboration, broadens students' educational horizons, and enhances their ability to innovate and creatively solve complex problems.

On the other hand, design thinking facilitates deep engagement with specific engineering disciplines, enhancing students' mastery of disciplinary knowledge. This approach was supported by the research of Lai et al. (2022), who note that design thinking effectively enhances student performance and professional success in specialised courses. Design thinking promotes knowledge transfer, crucial for educational advancement (Malele and Ramaboka 2020; Melles, Howard, and Thompson-Whiteside 2012). Empirical studies, including those by Luna, Talavera, and Chong (2018), showed that it improves students' understanding of physics and its application in real-world scenarios. Additionally, design thinking engages students in exploratory learning, increasing academic productivity. For example, Lur, Cheah, and Lee (2022) reported improved design skills in physics education through deeper understanding of the subject's principles. Research by Tiwari and Rathore (2022), Rak (2023), and Ulibarri et al. (2014) indicated that design thinking facilitates the visualisation and comprehension of system requirements, effectively achieving various educational objectives. These studies suggest that design thinking can significantly enhance student learning outcomes.

4.4.1.2 Emotional needs. Design thinking offers tools and strategies that can create immersive and engaging learning experiences, crucial for student success and broader well-being. According to the Programme for International Student Assessment (PISA), engagement is a critical marker for assessing educational quality (Fredricks, Filsecker, and Lawson 2016). Design thinking courses not only enhance technical knowledge but also meet psychological needs and stimulate interest through its learning strategies. This approach increases students' effort and engagement, creating an environment conducive to independent exploration and proactive learning. Empirical evidence suggested that design thinking enhances students' self-efficacy, engagement, and motivation

Table 8. Capacity focus.

Capacity Focus	Stakeholder groups	Codes
	For Students	Knowledge Acquisition
		Emotional Needs
	For Educators	Educator Training & Support
		Teaching Tool Development
	For Society	Social Justic
		SDGs

compared to traditional instructional methods (Borgford-Parnell, Deibel, and Atman 2010; Liu, Lee, and Huang 2023) which increase agency for problem solving.

Curricula infused with design thinking not only increase self-efficacy towards innovation and technological knowledge (Rao, Koola, and Mehta 2018), but also address psychological needs and stimulate interest through active learning strategies (Borgford-Parnell, Deibel, and Atman 2010; Mentzer et al. 2023). Furthermore, studies reported an uplift in student effort and engagement (Dyer, Grey, and Kinnane 2017; Ulibarri et al. 2014), and design thinking reforms by Palacin-Silva et al. (2017) have been shown to increase motivation among science and engineering students. Goudarz Poursharif et al. (2021) 's case study emphasises that design thinking is a key factor in developing essential skills and boosting student confidence. Among the studies reviewed, only a few focused specifically on female students. Suligoj, Zavbi, and Avsec (2020)'s research indicates that female students can enhance team learning and decision-making capabilities. Chen, Cheng, and Wu (2023) 's findings showed a positive correlation between female students' creative self-efficacy and their motivation in design thinking courses, suggesting significant improvements in confidence and active participation among these students. These insights emphasise the importance of targeted interventions to support women in STEM fields.

4.4.2 Educator level

The International Society for Technology in Education (ISTE) Standards for Educators suggest that 'designer' will become increasingly important for educational professionals in the future (International Society for Technology in Education 2017). In the context of the deep integration of intelligent technologies and education, educational professionals play a pivotal role in this paradigm shift. They are not only key drivers of this engineering transformation but also central to the development and application of design thinking in both instructional design and assessment technologies (Arastoopour Irgens et al. 2015). When individuals in education clearly understand their own professional identity and recognise design thinking as a critical component of innovative practice, they can become powerful agents of educational change.

4.4.2.1 Educator training & support. When educators clearly understand their roles and recognise design thinking as a distinct component of innovative action, they become potent catalysts for educational transformation. Dym et al. (2005) emphasised the importance of fostering creative thinking abilities among education leaders, educational managers, and curriculum developers. The studies included in this review highlight that enhancing educator training and support through design thinking involves providing comprehensive and continuous assistance to refine their skills, teaching methodologies, and overall effectiveness (Henriksen, Mehta, and Mehta 2019; Koh et al. 2015). Brink et al. (2021) endorsed using design thinking as a modelling tool, alongside implementing teacher training and development programmes. To address real-world challenges, Henriksen, Mehta, and Mehta (2019) developed a design thinking course specifically for graduate-level instructors, equipping educators with the necessary skills to apply design thinking in their practice.

4.4.2.2 Teaching tool development. In educational institutions, the integration of design thinking into the development of teaching and learning tools involves incorporating information technology tools and techniques. This integration aims to create effective tools that enhance student learning and improve overall teaching outcomes (Wu, Hu, and Wang 2019). The widespread application of design thinking in the business sector provides educators with a valuable blueprint for shaping and delivering educational content (Kimbell 2011). As highlighted in the study of Henriksen, Mehta, and Mehta (2019), integrating design thinking into pedagogy can yield more comprehensive and impactful projects that enhance higher engineering education. For example, the development of online learning platforms has been significantly advanced by design-based research approaches (Srisawat and Wannapiroon 2022). McAlpine, Reidsema, and Allen (2006) and Rahman et al. (2019) highlighted powerful testing platforms for research in design thinking, encompassing areas like

search strategies, design decision-making, AI in design, and design cognition. Shah, Smith, and Woodward (2009) introduced procedures and principles for developing and validating measurement tools for design skills. Coleman et al. (2020) developed and validated such tools. Srisawat and Wannapiroon (2022) created a platform that combines an experiential design thinking process to assess educators' digital competencies. Galleguillos, Santelices, and Bustos (2019) used game-based methodologies to develop human-centred innovation approaches. Souza et al. (2020) argued that platforms like DTA4RE are crucial for using online resources to understand and develop design thinking principles and related technologies.

4.4.3 For society

The application of design thinking has the capacity to extend beyond technical domains and tackle important social issues like sustainable development (Dotson et al. 2020; Levine, Lesniewski, and Agogino 2016) and poverty reduction (Dyer, Grey, and Kinnane 2017). Its incorporation into higher engineering curricula makes it easier for students to apply abstract ideas to real-world problems, giving them the opportunity to address pressing social concerns. Adopting design thinking makes higher engineering education a more supportive and inclusive area that advances the growth of communities and society, particularly within the dimensions of sustainable development goals and social justice, spanning multifaceted dimensions.

4.4.3.1 Sustainability development goals. The design thinking process contributes to achieving the Sustainable Development Goals (SDGs) by fostering innovative solutions that are both sustainable and user centric. Researchers like Goudarz Poursharif et al. (2021) and Khanduja, Bihola, and Dave (2017) emphasised the importance of embedding sustainability, teamwork, and a focus on user-centred design into the development process. Delaney stressed the necessity for graduate students (2023b) and engineers (2023a) to engage in teamwork and maintain a strong focus on user-centred designs. This approach not only enhances the functionality and acceptability of engineered solutions but also ensures that these solutions contribute to sustainable development by addressing specific societal and environmental challenges. This paradigm shift is essential in educating the next generation of engineers and designers to think broadly about the impacts of their work on both people and the planet. By emphasising these elements, design thinking encourages the creation of solutions that consider environmental impact, social equity, and economic viability, aligning with several SDGs including responsible consumption and production, climate action, and sustainable cities and communities (Dotson et al. 2020; Ping, Chow, and Teoh 2011). The application of design thinking can facilitate the development of more sustainable solutions by fostering a greater understanding of human needs in social and environmental contexts. Clark, Stabryla, and Gilbertson (2018) argued that employing design thinking to enhance sustainable design can challenge existing sustainability mindsets (Lande and Leifer 2010). Similarly, Alok and Saipriya (2020) and Charosky and Bragos (2021) advocated for leveraging design thinking to address global human challenges, highlighting its potential to generate innovative solutions for pressing issues.

4.4.3.2 Social justice. Social justice in education emphasises ensuring equitable access to high-quality learning opportunities for all individuals, regardless of background, socioeconomic status, or other factors (Latorre-Cosculluela et al. 2020; Leicht-Scholten and Steuer-Dankert 2020; Reynante 2022). Design thinking courses aim to promote greater diversity, equity, and inclusion (DEI) (Buzzanell, Eddington, and Zoltowski 2023) by addressing diverse cultural perspectives and working to eliminate systemic inequalities in engineering (Shetty and Xu 2018; Steuer-Dankert et al. 2019).

By promoting equal opportunities, these initiatives work to eliminate barriers such as socioeconomic and geographical disparities, as well as discrimination based on race, gender, or other factors (North 2008; Ulibarri et al. 2014). Research underscored the link between students' experiences of discrimination and their academic performance, highlighting the need for educational

institutions to actively foster social justice (Reichenbach and Reynante 2019), which is pivotal in enhancing student achievement.

Rak (2023) focused on leveraging design thinking to address issues of inclusion and marginalisation within higher engineering education. Additionally, Chuah (2021) found that game design thinking motivates rural students and provides those in remote areas with more interactive and practical learning experiences, thereby improving their opportunities in higher engineering education. This approach helped to mitigate resource inequality and ensures all students have equal opportunities to excel and contribute meaningfully to the field (Howe-Walsh and Turnbull 2016). Furthermore, fostering engineering identities among diverse student populations, is another key benefit of this educational strategy (Rodriguez et al. 2020).

5. Discussion

5.1. Interpreting the current landscape

The substantial research growth in recent years signals design thinking's emergence as a significant pedagogical movement in engineering education. However, interpreting this landscape requires acknowledging what this review set out to examine. As Crilly (2024) observes numerous 'disciplinary approaches' are described 'in isolation from one another', making relationships between them difficult to establish. This fragmentation motivated our study of 'design thinking' as a specific, explicitly labelled pedagogical movement rather than attempting to synthesise the entire terrain of design-related pedagogy in engineering. It notes that complaints about 'the ambiguity of what Design Thinking means' are not unique failures but rather 'characteristic of all such projects' that attempt to codify disciplinary essence. Our findings confirm this observation. Beneath the research growth lies a productive paradox where convergence in conceptual elements coexists with fragmentation in implementation. The dominance of Brown (2008), Dym et al. (2005) and Buchanan (1992) as definitional sources, alongside widespread adoption of the Stanford five-stage model, suggests shared conceptual foundations. Yet course-level implementation predominates over programme-wide integration, revealing that design thinking typically enters curricula through individual faculty initiatives rather than institutional transformation. This creates adaptability but risks incoherence as implementations vary widely without explicit rationales. Our review examined this phenomenon through the conceptual distinction, Crilly (2024) articulates between narrowly defined, process-oriented 'design thinking' (typically associated with IDEO and Stanford d.school frameworks) and the broader theoretical domain of 'designerly ways of knowing, thinking, and acting' (Cross 1982). The diversity in how 'design thinking' is defined and operationalised represents not conceptual failure but rather the empirical reality of how an educational movement adapts across diverse institutional contexts while maintaining identifiable core commitments.

The methodological pluralism observed reflects design thinking's complexity encompassing cognitive, affective, and behavioural dimensions. However, significant gaps exist. While surveys and project rubrics are commonly employed, direct observation remains rare, meaning the evidence base relies heavily on self-reports rather than process observation. The four core competencies consistently identified (creativity, collaboration, empathy, and problem-solving) demonstrate consensus around what design thinking should develop, yet how these competencies interconnect remains underexplored. The emphasis on collaboration aligns with Van Helden et al. (2023) findings, though our review reveals collaboration is specifically framed within design thinking contexts through empathy-building, interdisciplinary teamwork, and iterative critique.

The literature emphasises process frameworks, mindsets, and individual competencies more than epistemic tools, distributed cognition, or socio-material practice. This possibly reflects dominance of cognitivist frameworks locating thinking primarily as internal mental processes. In contrast, pragmatist perspectives (Mitcham 1994) view thinking as occurring through action and technology, while

post-cognitive perspectives view thinking as distributed across minds, bodies, tools, and social contexts.

This theoretical orientation has implications for disciplinary knowledge's underemphasised role. Our review reveals asymmetry: while many studies position design thinking as enhancing disciplinary knowledge through applied contexts, few investigate how students draw upon disciplinary knowledge during design. Yet Samuel and Lewis (1991) showed even simple tasks required 96 concepts, and research demonstrates students struggle to connect engineering principles during design (Carberry and McKenna 2014; Eastman, Newstetter, and McCracken 2001). Wolmarans (2016) demonstrated that successful teams can utilise disciplinary knowledge as epistemic tool, what Markauskaite and Goodyear (2017) term 'epistemic fluency', but such integration requires explicit scaffolding. This emphasis on design thinking as context for applying knowledge rather than examining how disciplinary knowledge enables design may reflect the movement's promotional origins. Evidence strength varies substantially across adoption rationales. Strongest support exists for immediate affective outcomes (engagement, motivation, self-efficacy), with moderate evidence for competency development but weakest evidence for long-term impacts given scarce longitudinal follow-up. Faculty training receives limited attention despite implementation requiring substantial pedagogical expertise, and explicit examination of equity dimensions remains rare despite persistent underrepresentation in engineering. This evidence gradient reveals we know considerably more about design thinking's immediate effects than its ultimate impacts on professional practice or career trajectories.

5.2. Recommendations

Given the broad range of topics covered by these studies, several recommendations emerge:

5.2.1 Focus on long-term changes and impact

Design thinking is often introduced into engineering education through separate modules in the curriculum, which limits the potential impact on student skills. Future research should continue exploring best practices for embedding design thinking into engineering curricula beyond individual projects and examining the long-term impacts on student outcomes and societal contributions.

5.2.2 Detailed steps for design thinking

The lack of a standardised definition of design thinking is a severe limitation in integration, as well as the lack of clarity between design thinking mindsets and design thinking processes. Future research should develop a detailed step-by-step syllabus based on this classic design thinking model but optimised towards engineering education. Potential characteristics of such a syllabus would be ease of implementation, impact on skills and industrial relevance. This may require synthesising existing models and providing structured approaches.

5.2.3 Policy and standards development

Educational standards set by accrediting agencies often include the requirement for design courses in engineering education. The fourth edition of 'Accreditation of Higher Education Programmes' (AHEP), established in August 2020 in line with the UK Standard for Professional Engineering Competence, describes engineering design as the process of developing and improving products, processes, or systems that are economically viable and meet specific needs, emphasising the significant level of intellectual and technical effort required at each stage of academic development. There is a growing demand for engineers in the UK and Chartered Engineers are expected to be not only technically proficient, but also to have clear design skills to deal with the complexity of modern engineering challenges (Council 2024). The absence of a standardised framework results in varied interpretations and applications of design thinking across educational institutions, which can lead to inconsistencies in student learning outcomes. Clear policies can provide a unified direction and

set expectations for both educators and students, ensuring that design thinking principles are uniformly integrated into engineering curricula.

5.2.4 Addressing drawbacks and negative impacts

While the report highlights positive aspects, it is important to recognise potential conflicts and frustrations arising from diverse professional backgrounds, which can reduce efficiency (Levy 2017). To address this, future research should focus on optimising resource utilisation, such as teaching staff and specific workspaces. It is also essential to determine how to allocate expertise in teaching and innovation skills within limited timeframes and resources (Charosky et al. 2022). Additionally, conducting disciplinary assessments is another important issue that requires attention. Teaching design thinking should not only incorporate a systematic learning design from the learners' perspective but also provide clear guidance for learning activities that embody the essence of design thinking. Shah, Smith, and Woodward (2009) suggested further research on the assessment and evaluation of design thinking skills.

5.2.5 Focus on underrepresented groups

Increasing awareness in higher engineering education has highlighted the varying demands of different student groups. The persistent underrepresentation of women in engineering fields remains a significant issue that has garnered considerable attention recently (Powell, Bagilhole, and Dainty 2009). The increasing number of female students departing from STEM degree programmes at universities has exacerbated this problem (Meiksins et al. 2013). Ensuring engagement and retention is particularly vital for students who are underrepresented in engineering. Targeted actions are necessary to bridge the gender gap in engineering education, promote diversity and inclusivity, and thereby foster a more diverse and inclusive environment (Eddington et al. 2020; Leicht-Scholten and Steuer-Dankert 2020). This emphasis on inclusivity is crucial for creating educational systems that not only accommodate but also actively support the varied needs of all students. Furthermore, equity discussions must include students with disabilities, a group often overlooked in current research. Most studies focus on students or makerspace participants, rather than faculty, staff, or makerspace facilitators. Addressing these gaps is essential for fostering an educational environment that supports the diverse needs of all individuals within the academic community.

6. Conclusions

This study investigates the major trends, methodologies, and emerging themes of contemporary design thinking application in higher engineering education. By searching the literature from 2003 to 2023 using three research databases and conference proceedings, supplemented by snowball sampling, we identified 76 relevant papers. This discussion expands on the results of each research question and proposes future directions for early engineering education research. Our findings offer valuable guidance for educators and researchers aiming to adapt their approaches to meet the evolving needs of learners.

The field demonstrates substantial growth in recent years, indicating widespread adoption across diverse institutional contexts. Design thinking has emerged as a significant pedagogical movement addressing perceived gaps in traditional engineering education, particularly regarding development of synthetic thinking, creative capabilities, collaborative skills, and human-centred approaches essential for contemporary engineering practice. However, most approaches are still in planning phase, with implementations often through separate modules limiting potential impact on student skills.

Implementation predominantly occurs through course-level integration rather than systematic programme-wide approaches, with semester-long courses and intensive workshops as primary formats. This pattern suggests design thinking often enters curricula through individual faculty

champions rather than institutional mandates, creating both adaptability and fragmentation. Conceptual foundations demonstrate convergence around key definitional sources and widespread adoption of the Stanford five-stage model, though frequently adapted through condensation, extension, or integration with established frameworks. Four core competencies consistently emerge: creativity, collaboration, empathy, and problem-solving. However, this convergence coexists with conceptual ambiguity as design thinking simultaneously functions as process model, mindset orientation, competency framework, and problem-solving methodology.

Adoption rationales operate at the student level, including developing professional competencies and fostering innovation mindsets, educator level including meeting accreditation requirements, and at the societal level, including addressing sustainability and social justice. Evidence strength varies substantially, with strongest support for immediate affective outcomes, moderate evidence for competency development, but the weakest evidence for long-term impacts, with few conducting longitudinal follow-up. This evidence gradient reveals that we know considerably more about design thinking's immediate effects than about ultimate impacts on professional practice or career trajectories.

6.1. Limitations

The current research has a few limitations that warrant acknowledgment. The primary limitation of this review lies in our deliberate focus on literature that explicitly uses the term 'design thinking'. This methodological choice, while aligned with our aim to study design thinking as a specific educational movement with distinct characteristics and institutional implementations, necessarily excludes important research on related concepts.

Specifically, our review does not capture 'design cognition' research exploring cognitive processes in design activities, 'design process' and 'design heuristics' studies examining procedural and strategic aspects of design, research on 'designerly ways of knowing' exploring epistemological foundations of design, or general 'engineering design education' literature that may address similar pedagogical challenges using different terminology. While these literatures offer valuable insights into the broader landscape of design in engineering education, our focus on the specific term 'design thinking' enabled us to examine how this educational movement – with its distinctive frameworks, terminology, promotional infrastructure, and institutional implementations – has been conceptualised and practiced in engineering education contexts.

This narrow scope has important implications for how our findings should be interpreted. Most notably, relevant work engaging deeply with design thinking principles but using different terminology falls outside our review. Studies examining thinking-through-tools, distributed cognition in design, or reflective practice in engineering may address similar phenomena without adopting the 'design thinking' label. The design cognition and designerly ways of knowing traditions contain theoretical frameworks and empirical insights that could enrich understanding of design thinking practice, yet our review captures what the 'design thinking' movement says about itself rather than what related scholarly traditions might say about the same phenomena.

This affects generalisability in important ways. Our findings pertain specifically to the 'design thinking' movement as explicitly labelled and should not be overgeneralised to design education broadly or to the broader intellectual tradition of 'designerly ways of knowing'. Engineering educators and researchers interested in design pedagogy more generally should consult reviews with broader scope, such as Van Helden et al. (2023) on collaborative learning in engineering design education.

Our scope also reflects movement boundaries as self-identified rather than conceptual boundaries scholars might draw based on substantive analysis. This is appropriate for understanding a social phenomenon.– the adoption of 'design thinking' as a reform movement in engineering education,– but it means we are examining one discourse community's practices rather than comprehensively mapping all design-related pedagogy. We acknowledge this fundamental limitation and

emphasise that our conclusions pertain to 'design thinking' as an explicitly labelled pedagogical movement rather than to design education or designerly thinking broadly.

Future research employing broader search strategies that include synonymous terminology could provide complementary insights into the wider terrain of design education in engineering.

6.2. Future directions

Design thinking will play a crucial role in the development of innovative and non-conventional learning resources. The landscape of engineering education is undergoing significant changes due to the commoditisation of engineering and basic science knowledge and the increasing adoption of blended learning approaches (Chesler et al. 2015). Future research and practice should focus on integrating advanced technologies with design thinking frameworks to further personalise and enrich engineering education. By doing so, educators can create dynamic, interactive, and inclusive learning environments that better prepare students to tackle the challenges of the modern world. Advanced technologies combined with design thinking frameworks can help create personalised and inclusive learning environments. Personalised learning paths ensure that each student's unique needs and learning styles are addressed, thereby improving engagement and retention. Inclusive design practices ensure that educational resources are accessible to all students, including those with disabilities. By prioritising personalisation and inclusivity, educators can ensure that all students succeed in engineering fields.

The combination of artificial intelligence (AI) and design thinking is poised to revolutionise educational practices. AI can provide personalised learning experiences by analysing student data to tailor educational content and feedback to individual needs. Integrating AI with design thinking can enhance the problem-solving process, offering intelligent suggestions and real-time insights that guide students through iterative design cycles. This integration can foster a more adaptive and responsive learning environment, promoting deeper understanding and engagement.

The shift towards blended learning approaches, which combine traditional face-to-face instruction with online learning, requires the development of robust digital resources. Utilising online platforms to understand design thinking principles and related technologies is crucial (Souza et al. 2020). Digital resources can offer interactive and immersive experiences that are not possible in a conventional classroom setting. For instance, virtual and augmented reality tools can simulate real-world scenarios, providing students with hands-on experience in a controlled environment. McAlpine, Reidsema, and Allen (2006) emphasised the importance of digital resources in enhancing the learning.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Additional information

The authors received no financial support for the research, authorship, and/or publication of this article.

Acknowledgements

The authors acknowledge the use of ChatGPT (OpenAI, GPT-4 version prior to version 5.2) during the revision process of this manuscript. The tool was used solely to refine language clarity, improve grammatical accuracy, and enhance readability of the text. All substantive content, including research design, literature search and screening, data extraction, coding and analysis, framework development, interpretation of findings, and conclusions, remains entirely the work of the authors. ChatGPT was not involved in any analytical decisions, methodological choices, or the synthesis of results.

The authors have carefully reviewed and verified all content and take full responsibility for the accuracy and integrity of this manuscript.

Notes on contributors

Yuwei Deng is a PhD candidate in the Department of Engineering at King's College London. Her current research examines how design principles can be integrated into engineering curricula to enhance both teaching approaches and student learning outcomes. Her work explores how students develop designerly capabilities, how educators teach design competencies, and how institutions create conditions for design-centred learning across diverse contexts, with the aim of equipping future engineers with creative problem-solving skills for complex challenges. She is broadly interested in the intersection of design and engineering education, and how this integration can transform traditional curricula.

Wei Liu is Professor of Design Engineering and Innovation at King's College London and Associate Vice Dean in the Faculty of Natural, Mathematical and Engineering Sciences. She holds a PhD from the University of Cambridge and was a Visiting Fellow at Harvard University. Prior to joining King's, she held a range of academic and leadership roles across the UK, US, and China. Professor Liu is an elected Member of the European Academy of Sciences and Arts, a Fellow of the Royal Society for the Encouragement of Arts, Manufactures and Commerce (FRSA), and a Senior Fellow of the Higher Education Academy (SFHEA).

Claire Lucas was Professor of Engineering Education & Skills and Deputy Head (Education) in the Department of Engineering at King's College London. She is currently Executive Dean and Professor of Engineering Education & Skills at Aston University. Her research focuses on the behaviours, skills, and curricula required of future engineers, alongside the application of systems, quality, and design frameworks to enhance engineering education. She supervised the creation of the UK's first behavioural taxonomy for engineering and introduced the key-shaped engineer model. She contributes to national standards through roles with the QAA, the Royal Academy of Engineering's Education & Skills Committee, the Engineering Council Registration Standards Committee, and the IET Academic Accreditation Committee.

References

Açar, A., and D. Rother. 2011. "Design Thinking in Engineering Education and Its Adoption in Technology-Driven Startups." Advances in Sustainable Manufacturing: Proceedings of the 8th Global Conference on Sustainable Manufacturing.

Acebo, E., J. Miguel-Dávila, and L. Herrera. 2021. "Design Thinking (DT) in Engineering Education (EE): A Systematic Literature Review (SLR)." International Conference on Industrial Engineering and Operations Management.

Aguilera, D., and J. Ortiz-Revilla. 2021. "STEM vs. STEAM Education and Student Creativity: A Systematic Literature Review." *Education Sciences* 11 (7): 331. https://doi.org/10.3390/educsci11070331

Alok, G., and P. Saipriya. 2020. "A Corroborative Approach for Engineering Education Using Design Thinking." *Journal of Engineering Education Transformations* 33:429–433. https://doi.org/10.16920/jeet/2020/v33i0/150195

Al-Qaralleh, E., B. H. Sababha, and K. Abugharbieh. 2021, May 17-18. "Integrating Design Thinking in Freshmen-Level Engineering Curriculum." [Proceedings of the 2021 Innovation and new Trends in Engineering, Science and Technology Education Conference (ietsec 2021)]. Innovation and New Trends in Engineering, Technology and Science Education Conference (IETSEC), Electr Network.

Arastoopour Irgens, G., D. W. Shaffer, Z. Swiecki, A. Ruis, and N. C. Chesler. 2015. "Teaching and Assessing Engineering Design Thinking with Virtual Internships and Epistemic Network Analysis." *International Journal of Engineering Education* 32: 1492–1501.

Archer, B. 1979. "Design as a Discipline." *Design Studies* 1 (1): 17–20. https://doi.org/10.1016/0142-694X(79)90023-1

Atman, C. J., R. S. Adams, M. E. Cardella, J. Turns, S. Mosborg, and J. Saleem. 2007. "Engineering Design Processes: A Comparison of Students and Expert Practitioners." *Journal of Engineering Education* 96 (4): 359–379. https://doi.org/10.1002/j.2168-9830.2007.tb00945.x

Bairaktarova, D., and D. Plumlee. 2022. "Creating Space for Empathy: Perspectives on Challenges of Teaching Design Thinking to Future Engineers [Article]." *International Journal of Engineering Education* 38 (2): 512–524. <Go to ISI>://WOS:000758662400020.

Berggren, K.-F., D. Brodeur, E. F. Crawley, I. Ingemarsson, W. T. Litant, J. Malmqvist, and S. Östlund. 2003. "CDIO: An International Initiative for Reforming Engineering Education." *World Transactions on Engineering and Technology Education* 2 (1): 49–52.

Bhatnagar, T., and P. Badke-Schaub. 2017, Jan 09-11. Design Thinking and Creative Problem Solving for Undergraduate Engineering Education in India: The Need and Relevance. *Smart Innovation Systems and Technologies* [Research into Design for Communities, vol 2: Proceedings of icord 2017]. 6th International Conference on Research into Design (ICoRD), Indian Inst Technol, Guwahati, INDIA.

Borgford-Parnell, J., K. Deibel, and C. J. Atman. 2010. "From Engineering Design Research to Engineering Pedagogy: Bringing Research Results Directly to the Students [Article]." *International Journal of Engineering Education* 26 (4): 748–759. <Go to ISI>://WOS:000282010500003.

Bork, S. J., and J. L. Mondisa. 2022. "Engineering Graduate Students' Mental Health: A Scoping Literature Review." *Journal of Engineering Education* 111 (3): 665–702. https://doi.org/10.1002/jee.20465

Borrego, M., M. J. Foster, and J. E. Froyd. 2014. "Systematic Literature Reviews in Engineering Education and Other Developing Interdisciplinary Fields." *Journal of Engineering Education* 103 (1): 45–76. https://doi.org/10.1002/jee.20038

Bosman, L. 2019. "From Doing to Thinking: Developing the Entrepreneurial Mindset through Scaffold Assignments and Self-regulated Learning Reflection." *Open Education Studies* 1 (1): 106–121. https://doi.org/10.1515/edu-2019-0007

Braun, V., and V. Clarke. 2006. "Using Thematic Analysis in Psychology." *Qualitative Research in Psychology* 3 (2): 77–101. https://doi.org/10.1191/1478088706qp063oa

Brenner, W., F. Uebernickel, and T. Abrell. 2016. *Design Thinking for Innovation: Research and Practice*. Cham: Springer International Publishing.

Brereton, M. 2004. "Distributed Cognition in Engineering Design: Negotiating between Abstract and Material Representations." In *Design Representation*, edited by G. Goldschmidt and W. L. Porter, 83–103. London: Springer.

Brink, S., C. J. Carlsson, M. Enelund, F. Georgsson, E. Keller, R. Lyng, C. McCartan, and Ieee. 2021, Oct 13-16. Curriculum Agility: Responsive Organization, Dynamic Content, and Flexible Education. [2021 Ieee Frontiers in Education Conference (fie 2021)]. IEEE Frontiers in Education Conference (FIE), Lincoln, NE.

Brown, T. 2008. "Design Thinking." *Harvard Business Review* 86 (6): 84.

Brown, T., and B. Katz. 2009. *Change by Design*. New York: HaperCollinns.

Bucciarelli, L. L. 2001. "Design Knowing & Learning: A Socially Mediated Activity." In *Design Knowing and Learning: Cognition in Design Education*, edited by C. M. Eastman, W. M. McCracken, and W. C. Newstetter, 297–314. Amsterdam: Elsevier.

Buchanan, R. 1992. "Wicked Problems in Design Thinking." *Design issues* 8 (2): 5–21. https://doi.org/10.2307/1511637

Buzzanell, P. M., S. M. Eddington, and C. B. Zoltowski. 2023. "Facilitation for Diversity, Equity, and Inclusion through Design Thinking." 2023 ASEE Annual Conference & Exposition.

Camacho, M. 2016. "David Kelley: From Design to Design Thinking at Stanford and IDEO." *She Ji: The Journal of Design, Economics, and Innovation* 2 (1): 88–101.

Carberry, A. R., and A. F. McKenna. 2014. "Exploring Student Conceptions of Modeling and Modeling Uses in Engineering Design." *Journal of Engineering Education* 103 (1): 77–91. https://doi.org/10.1002/jee.20033

Charosky, G., and R. Bragos. 2021. "Investigating Students' Self-perception of Innovation Competences in Challenge-Based and Product Development Courses [Article]." *International Journal of Engineering Education* 37 (2): 461–470. <Go to ISI>://WOS:000641026600014.

Charosky, G., L. Hassi, L. Leveratto, K. Papageorgiou, J. Ramos, and R. Bragos. 2018. Education for Innovation: Engineering, Management and Design Multidisciplinary Teams of Students Tackling Complex Societal Problems through Design Thinking. 4th International Conference on Higher Education Advances (HEAD'18).

Charosky, G., L. Hassi, K. Papageorgiou, and R. Bragos. 2022. "Developing Innovation Competences in Engineering Students: A Comparison of two Approaches [Article]." *European Journal of Engineering Education* 47 (2): 353–372. https://doi.org/10.1080/03043797.2021.1968347.

Chen, M.-H., S.-P. Cheng, and L. Y. Wu. 2023. "Merging Design Thinking into Translational Research in a Biomedical Engineering Laboratory (DT-TRBEL) Course." *Sustainability* 15 (18): 13688. https://doi.org/10.3390/su151813688

Chesler, N. C., A. R. Ruis, W. Collier, Z. Swiecki, G. Arastoopour, and D. W. Shaffer. 2015. "A Novel Paradigm for Engineering Education: Virtual Internships with Individualized Mentoring and Assessment of Engineering Thinking [Article]." *Journal of Biomechanical Engineering-Transactions of the Asme* 137 (2): 024701. https://doi.org/10.1115/1.4029235.

Chuah, K. M., F. S. Mohamad, J. L. Minoi, M. K. Kabilan, and Ieee. 2021, Nov 15-18. Of Marbles and Minecraft: Designing STEM Educational Games for Culturally Diverse Young Learners in Malaysia. [2021 World Engineering Education Forum/Global Engineering Deans Council (weef/gedc)]. World Engineering Education Forum (WEEF) / Global-Engineering-Deans-Council (GEDC) Conference, Madrid, SPAIN.

Clark, R. M., L. M. Stabryla, and L. M. Gilbertson. 2018. Use of Active Learning and the Design Thinking Process to Drive Creative Sustainable Engineering Design Solutions. 2018 ASEE Annual Conference & Exposition.

Clark, R. M., L. M. Stabryla, and L. M. Gilbertson. 2020. "Sustainability Coursework: Student Perspectives and Reflections on Design Thinking [Article]." *International Journal of Sustainability in Higher Education* 21 (3): 593–611. https://doi.org/10.1108/ijshe-09-2019-0275.

Cohen, J. 1960. "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20 (1): 37–46. https://doi.org/10.1177/001316446002000104

Coleman, E., T. Shealy, J. Grohs, and A. Godwin. 2020. "Design Thinking among First-Year and Senior Engineering Students: A Cross-Sectional, National Study Measuring Perceived Ability." *Journal of Engineering Education* 109 (1): 72–87. https://doi.org/10.1002/jee.20298

Colmenares-Quintero, R. F., D. M. Caicedo-Concha, N. Rojas, K. E. Stansfield, and J. C. Colmenares-Quintero. 2023. "Problem Based Learning and Design Thinking Methodologies for Teaching Renewable Energy in Engineering Programs: Implementation in a Colombian University Context." *Cogent Engineering* 10 (1): 2164442. https://doi.org/10.1080/23311916.2022.2164442

Cooper, B. 2011. *Empathy in Education: Engagement, Values and Achievement*. London: Bloomsbury Publishing.

Corral, L., and I. Fronza. 2018. "Design Thinking and Agile Practices for Software Engineering: An Opportunity for Innovation." Proceedings of the 19th Annual SIG Conference on Information Technology Education.

Council, E. C. E. C. 2024.

Crawley, E., J. Malmqvist, S. Ostlund, D. Brodeur, and K. Edstrom. 2007. "Rethinking Engineering Education." *The CDIO Approach* 302 (2): 60–62.

Crawley, E. F., J. Malmqvist, S. Östlund, and D. R. Brodeur. 2007. *Rethinking Engineering Education: The CDIO Approach*. New York: Springer.

Crilly, N. 2024. *Design Thinking and Other Approaches: How Different Disciplines See, Think and Act*. Cambridge: Cambridge University Press.

Cross, N. 1982. "Designerly Ways of Knowing." *Design Studies* 3 (4): 221–227. https://doi.org/10.1016/0142-694X(82)90040-0

Daly, S. R., R. S. Adams, and G. M. Bodner. 2012. "What Does It Mean to Design? A Qualitative Investigation of Design Professionals' Experiences." *Journal of Engineering Education* 101 (2): 187–219. https://doi.org/10.1002/j.2168-9830.2012.tb00048.x

Davies, P. 2000. "The Relevance of Systematic Reviews to Educational Policy and Practice." *Oxford Review of Education* 26 (3-4): 365–378. https://doi.org/10.1080/713688543

de Carvalho, C. V., Š Cerar, J. Rugelj, H. Tsalapatas, and O. Heidmann. 2020. "Addressing the Gender gap in Computer Programming through the Design and Development of Serious Games." *IEEE Revista Iberoamericana de Tecnologias del Aprendizaje* 15 (3): 242–251. https://doi.org/10.1109/RITA.2020.3008127

DeHaan, R. L. 2009. "Teaching Creativity and Inventive Problem Solving in Science." *CBE – Life Sciences Education* 8 (3): 172–181. https://doi.org/10.1187/cbe.08-12-0081

Delaney, E., and W. Liu. 2023a. "An Investigation into the Current Status and Importance of Sustainability Education for Designers and Engineers." *International Journal of Engineering Education* 39 (3): 502–518.

Delaney, E., and W. Liu. 2023b. "Postgraduate Design Education and Sustainability – an Investigation into the Current State of Higher Education and the Challenges of Educating for Sustainability." *Frontiers in Sustainability* 4:1148685. https://doi.org/10.3389/frsus.2023.1148685

Dell'Era, C., S. Magistretti, C. Cautela, R. Verganti, and F. Zurlo. 2020. "Four Kinds of Design Thinking: From Ideating to Making, Engaging, and Criticizing." *Creativity and Innovation Management* 29 (2): 324–344. https://doi.org/10.1111/caim.12353

de Paula, D., K. Cormican, and F. Dobrigkeit. 2021. "From Acquaintances to Partners in Innovation: An Analysis of 20 Years of Design Thinking's Contribution to new Product Development." *IEEE Transactions on Engineering Management* 69 (4): 1664–1677. https://doi.org/10.1109/TEM.2021.3084884

Dorst, K. 2011. "The Core of 'Design Thinking'and Its Application." *Design Studies* 32 (6): 521–532. https://doi.org/10.1016/j.destud.2011.07.006

Dotson, M. E., V. Alvarez, M. Tackett, G. Asturias, I. Leon, and N. Ramanujam. 2020. "Design Thinking-Based STEM Learning: Preliminary Results on Achieving Scale and Sustainability through the IGNITE Model [Article]." *Frontiers in Education* 5:10. Article 14. doi:10.3389feduc.2020.00014.

Dyer, M., T. Grey, and O. Kinnane. 2017. "GiveMe Shelter: A People-Centred Design Process for Promoting Independent Inquiry-led Learning in Engineering [Article]." *European Journal of Engineering Education* 42 (6): 729–744. https://doi.org/10.1080/03043797.2016.1219697.

Dym, C. L., A. M. Agogino, O. Eris, D. D. Frey, and L. J. Leifer. 2005. "Engineering Design Thinking, Teaching, and Learning [Review]." *Journal of Engineering Education* 94 (1): 103–120. https://doi.org/10.1002/j.2168-9830.2005.tb00832.x.

Dzombak, R., and S. Beckman. 2020. "Unpacking Capabilities Underlying Design (Thinking) Process [Article]." *International Journal of Engineering Education* 36 (2): 574–585. <Go to ISI>://WOS:000518888400007.

Eastman, C., W. Newstetter, and M. McCracken. 2001. *Design Knowing and Learning: Cognition in Design Education*. Amsterdam: Elsevier.

Eddington, S. M., D. Corple, P. M. Buzzanell, C. Zoltowski, and A. Brightman. 2020. "Addressing Organizational Cultural Conflicts in Engineering with Design Thinking [Article]." *Negotiation and Conflict Management Research* 13 (3): 263–284. https://doi.org/10.1111/ncmr.12191.

Edström, K., and A. Kolmos. 2014. "PBL and CDIO: Complementary Models for Engineering Education Development." *European Journal of Engineering Education* 39 (5): 539–555. https://doi.org/10.1080/03043797.2014.895703

Ericson, A., M. Bergstrom, A. C. Larsson, and P. Torlind. 2009a, Aug 24-27. Design Thinking Challenges in Education. [Iced 09 – The 17th International Conference on Engineering Design, vol 10: Design Education and Lifelong Learning]. 17th International Conference on Engineering Design, Stanford Univ, Stanford, CA.

Ericson, Å, M. Bergström, A. C. Larsson, and P. Törlind. 2009b. Design Thinking Challenges in Education. International Conference on Engineering Design: 24/08/2009-27/08/2009.

Fila, N. D., S. McKIlligan, and K. Guerin. 2018. Design Thinking in Engineering Course Design. 2018 ASEE Annual Conference & Exposition.

Finelli, C. J., M. Borrego, and G. Rasoulifar. 2015. "Development of a Taxonomy of Keywords for Engineering Education Research." *IEEE Transactions on Education* 58 (4): 219–241.

Fleiss, J. L., B. Levin, and M. C. Paik. 2013. *Statistical Methods for Rates and Proportions*. Hoboken: John Wiley & Sons.

Forest, C. R., R. A. Moore, A. S. Jariwala, B. B. Fasse, J. Linsey, W. Newstetter, P. Ngo, and C. Quintero. 2014. "The Invention Studio: A University Maker Space and Culture." *Advances in Engineering Education* 4 (2): Art. n2.

Fredricks, J. A., M. Filsecker, and M. A. Lawson. 2016. *Student Engagement, Context, and Adjustment: Addressing Definitional, Measurement, and Methodological Issues*. Vol. 43, 1–4. Amsterdam: Elsevier.

Galleguillos, L., I. Santelices, and R. Bustos. 2019. Designing a Board Game for Industrial Engineering Students. A Collaborative Work Experience of Freshmen. INTED2019 Proceedings.

Gaston, J. P., S. K. Guffey, and A. M. Rand. 2023. "Using Video and Written Reflection to Assess Second-Grade Students' Design Thinking and Conceptual Understanding in an Engineering and Design Challenge." *International Journal of Education in Mathematics, Science and Technology* 11 (4): 820–843. https://doi.org/10.46328/ijemst.2746

Goldschmidt, G., and P. A. Rodgers. 2013. "The Design Thinking Approaches of Three Different Groups of Designers Based on Self-reports." *Design Studies* 34 (4): 454–471. https://doi.org/10.1016/j.destud.2013.01.004

Golkar, A. 2020. "Experiential Systems Engineering Education Concept Using Stratospheric Balloon Missions [Article]." *Ieee Systems Journal* 14 (2): 1558–1567. https://doi.org/10.1109/jsyst.2019.2917823.

Gough, D., J. Thomas, and S. Oliver. 2012. "Clarifying Differences between Review Designs and Methods." *Systematic Reviews* 1:1–9. https://doi.org/10.1186/2046-4053-1-28

Gough, D., J. Thomas, and S. Oliver. 2017. *An Introduction to Systematic Reviews*. 2nd ed. London: Sage Publications.

Grönman, S., and E. Lindfors. 2021. "The Process Models of Design Thinking: A Literature Review and Consideration from the Perspective of Craft, Design and Technology Education." *Techne serien-Forskning I slöjdpedagogik och slöjdvetenskap* 28 (2): 110–118.

Guaman-Quintanilla, S., P. Everaert, K. Chiluiza, and M. Valcke. 2023. "Impact of Design Thinking in Higher Education: A Multi-actor Perspective on Problem Solving and Creativity." *International Journal of Technology and Design Education* 33 (1): 217–240. https://doi.org/10.1007/s10798-021-09724-z

Guerra, M., and T. Shealy. 2018. "Teaching User-Centered Design for More Sustainable Infrastructure through Role-Play and Experiential Learning [Article]." *Journal of Professional Issues in Engineering Education and Practice* 144 (4): 05018016. https://doi.org/10.1061/(asce)ei.1943-5541.0000385.

Henriksen, D., R. Mehta, and S. Mehta. 2019. "Design Thinking Gives STEAM to Teaching: A Framework That Breaks Disciplinary Boundaries." In *Steam Education: Theory and Practice*, edited by M. S. Khine and S. Areepattamannil, 57–78. Cham: Springer.

Howe-Walsh, L., and S. Turnbull. 2016. "Barriers to Women Leaders in Academia: Tales from Science and Technology." *Studies in Higher Education* 41 (3): 415–428. https://doi.org/10.1080/03075079.2014.929102

Ibwe, K., E. Kalinga, N. Mvungi, and H. Tenhunen. 2018, Jul 02-04. Effective Problems Solving Methods for Interdisciplinary Engineering Students in Challenge Driven Education. *EDULEARN Proceedings* [Edulearn18: 10th International Conference on Education and new Learning Technologies]. 10th International Conference on Education and New Learning Technologies (EDULEARN), Palma, SPAIN.

International Society for Technology in Education. 2017. *ISTE Standards for Educators*. International Society for Technology in Education. https://www.iste.org/standards/for-educators.

Isa, C. M. M., N. K. Mustaffa, C. N. Preece, and W.-K. Lee. 2019. Enhancing Conceive-Design-Implement-Operate and Design Thinking (CDIO-DT) Skills through Problem-Based Learning Innovation Projects. 2019 IEEE 11th International Conference on Engineering Education (ICEED),

Jiang, C., and Y. Pang. 2023. "Enhancing Design Thinking in Engineering Students with Project-Based Learning." *Computer Applications in Engineering Education* 31 (4): 814–830. https://doi.org/10.1002/cae.22608.

Juhl, J., and H. Lindegaard. 2013. "Representations and Visual Synthesis in Engineering Design." *Journal of Engineering Education* 102 (1): 20–50. https://doi.org/10.1002/jee.20001

Kaygan, P. 2023. "From Forming to Performing: Team Development for Enhancing Interdisciplinary Collaboration between Design and Engineering Students Using Design Thinking." *International Journal of Technology and Design Education* 33 (2): 457–478. https://doi.org/10.1007/s10798-022-09736-3

Khanduja, G. S., K. Bihola, and J. Dave. 2017, Jan 09-11. "Infusing Design Thinking Mindset in Engineering Education-A Pioneering Step towards Excellence and Sustainability." *Smart Innovation Systems and Technologies* [Research into Design for Communities, vol 2: Proceedings of icord 2017]. 6th International Conference on Research into Design (ICoRD), Indian Inst Technol, Guwahati, INDIA.

Kimbell, L. 2011. "Rethinking Design Thinking: Part I." *Design and Culture* 3 (3): 285–306. https://doi.org/10.2752/175470811X13071166525216

Kitchenham, B. 2004. "Procedures for Performing Systematic Reviews." *Keele, UK, Keele University* 33 (2004): 1–26.

Koh, J. H. L., C. S. Chai, B. Wong, and H.-Y. Hong. 2015. *Design Thinking and Education*. Singapore: Springer.

Kolodner, J. L. 2002. "Facilitating the Learning of Design Practices: Lessons Learned from an Inquiry into Science Education." *Journal of Industrial Teacher Education* 39 (3): 9–40.

Kolodner, J. L., J. Gray, and B. B. Fasse. 2003. "Promoting Transfer through Case-Based Reasoning: Rituals and Practices in Learning by Design Classrooms." *Cognitive Science Quarterly* 3 (2): 183–232.

Krippendorff, K. 2018. *Content Analysis: An Introduction to Its Methodology*. Thousand Oaks: Sage Publications.

Kruger, C., and N. Cross. 2006. "Solution Driven versus Problem Driven Design: Strategies and Outcomes." *Design Studies* 27 (5): 527–548. https://doi.org/10.1016/j.destud.2006.01.001

Kuo, H. C., Y. C. Tseng, and Y. T. C. Yang. 2019. "Promoting College Student's Learning Motivation and Creativity through a STEM Interdisciplinary PBL Human-Computer Interaction System Design and Development Course [Article]." *Thinking Skills and Creativity* 31:1–10. https://doi.org/10.1016/j.tsc.2018.09.001.

Kuo, H.-C., Y.-T. C. Yang, J.-S. Chen, T.-W. Hou, and M.-T. Ho. 2021. "The Impact of Design Thinking PBL Robot Course on College Students' Learning Motivation and Creative Thinking." *IEEE Transactions on Education* 65 (2): 124–131. https://doi.org/10.1109/TE.2021.3098295

Kuo, H. C., Y. T. C. Yang, J. S. Chen, T. W. Hou, and M. T. Ho. 2022. "The Impact of Design Thinking PBL Robot Course on College Students' Learning Motivation and Creative Thinking [Article]." *IEEE Transactions on Education* 65 (2): 124–131. https://doi.org/10.1109/te.2021.3098295.

Lai, C. F., H. X. Zhong, J. H. Chang, and P. S. Chiu. 2022. "Applying the DT-CDIO Engineering Design Model in a Flipped Learning Programming Course [Article]." *Etr&D-Educational Technology Research and Development* 70 (3): 823–847. https://doi.org/10.1007/s11423-022-10086-z.

Lande, M., and L. Leifer. 2009, Aug 24-27. Prototyping to Learn: Characterizing Engineering Students' Prototyping Activities and Prototypes. [Iced 09 – the 17th International Conference on Engineering Design, vol 1: Design Processes]. 17th International Conference on Engineering Design, Stanford Univ, Stanford, CA.

Lande, M., and L. Leifer. 2010. "Difficulties Student Engineers Face Designing the Future." *International Journal of Engineering Education* 26 (2): 271.

Landis, J. R., and G. G. Koch. 1977. "An Application of Hierarchical Kappa-Type Statistics in the Assessment of Majority Agreement among Multiple Observers." *Biometrics* 33 (2): 363–374. https://doi.org/10.2307/2529786.

Latorre-Cosculluela, C., S. Vázquez-Toledo, A. Rodríguez-Martínez, and M. Liesa-Orús. 2020. "Design Thinking: Creativity and Critical Thinking in College." *Revista electrónica de investigación educativa* 22: e22.

Lechuga, V. M., L. N. Clerc, and A. K. Howell. 2009. "Power, Privilege, and Learning: Facilitating Encountered Situations to Promote Social Justice." *Journal of College Student Development* 50 (2): 229–244. https://doi.org/10.1353/csd.0.0064

Lecy, J. D., and K. E. Beatty. 2012. Representative Literature Reviews Using Constrained Snowball Sampling and Citation Network Analysis. *Available at SSRN 1992601*.

Leem, S. Y., S. W. Lee, and ASME 2015, Aug 02-05. Interdisciplinary Design Education Based on Collaboration between Arts and Engineering Schools. [International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, 2015, vol 3]. ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, Boston, MA.

Leicht-Scholten, C., and L. Steuer-Dankert. 2020. "Educating Engineers for Socially Responsible Solutions through Design Thinking." In *Design Thinking in Higher Education: Interdisciplinary Encounters*, edited by T. Meinel and L. Leifer, 229–246. Singapore: Springer.

Levine, D. I., M. A. Lesniewski, and A. M. Agogino. 2016. "Design Thinking in Development Engineering [Article; Proceedings Paper]." *International Journal of Engineering Education* 32 (3): 1396–1406. <Go to ISI>:// WOS:000389146200010.

Levy, M. 2017, Jun 12-16. Design Thinking in Multidisciplinary Learning Teams: Insights from Multidisciplinary Teaching Events. *Lecture Notes in Business Information Processing* [Advanced Information Systems Engineering Workshops (Caise 2017)]. 29th International Conference on Advanced Information Systems Engineering (CAiSE), Essen, GERMANY.

Li, T., and Z. Zhan. 2022. "A Systematic Review on Design Thinking Integrated Learning in K-12 Education." *Applied Sciences* 12 (16): 8077. https://doi.org/10.3390/app12168077

Li, Y., A. H. Schoenfeld, A. A. diSessa, A. C. Graesser, L. C. Benson, L. D. English, and R. A. Duschl. 2019. "Design and Design Thinking in STEM Education." *Journal for STEM Education Research* 2: 93–104. https://doi.org/10.1007/s41979-019-00020-z.

Lin, K.-Y., Y.-T. Wu, Y.-T. Hsu, and P. J. Williams. 2021. "Effects of Infusing the Engineering Design Process into STEM Project-Based Learning to Develop Preservice Technology Teachers' Engineering Design Thinking." *International Journal of STEM Education* 8:1–15. https://doi.org/10.1186/s40594-020-00258-9

Liu, Y.-L. E., T.-P. Lee, and Y.-M. Huang. 2023. "Enhancing University Students' Creative Confidence, Learning Motivation, and Team Creative Performance in Design Thinking Using a Digital Visual Collaborative Environment." *Thinking Skills and Creativity* 50:101388. https://doi.org/10.1016/j.tsc.2023.101388

Lucas, B., G. Claxton, and J. Hanson. 2014. Thinking Like an Engineer: Implications for the Education System.

Luna, A., A. Talavera, M. Chong, and Ieee. 2018, Mar 11-14. How to Motivate the Interest in Physics to Engineering Students without Dying in the Attempt? [2018 Ieee World Engineering Education Conference (edunine)]. 2nd IEEE World Engineering Education Conference (EDUNINE), Univ CAECE, Buenos Aires, ARGENTINA.

Lur, K. T., C. W. Cheah, and C. H. Lee. 2022. Connecting Design and Engineering Physics with Reverse Engineering. 2022 IEEE Global Engineering Education Conference (EDUCON).

Mabogunje, A., N. Sonalkar, and L. Leifer. 2016. "Design Thinking: A new Foundational Science for Engineering." *International Journal of Engineering Education* 32 (3): 1540–1556.

Malele, V., and M. E. Ramaboka. 2020. "The Design Thinking Approach to Students STEAM Projects." *Procedia CIRP* 91:230–236. https://doi.org/10.1016/j.procir.2020.03.100.

Markauskaite, L., and P. Goodyear. 2017. *Innovation, Knowledgeable Action and Actionable Knowledge*. Dordrecht: Springer.

Mayring, P. 2014. Qualitative Content Analysis: Theoretical Foundation, Basic Procedures and Software Solution.

McAlpine, I., C. Reidsema, and B. Allen. 2006, Dec 03-06. Educational Design and Online Support for an Innovative Project-Based Course in Engineering Design. [Who's Learning? Whose Technology? Proceedings, vols 1 and 2]. 23rd Annual Conference of the Australasian-Society-for-Computers-in-Learning-in-Tertiary-Education, Univ Sydney, Ctr Res Comp Supported Learning & Cognit, Sydney, AUSTRALIA.

McDonald, J. K., R. E. West, P. J. Rich, and I. Pfleger. 2019. ""It's So Wonderful Having Different Majors Working Together": The Development of an Interdisciplinary Design Thinking Minor." *TechTrends* 63:440–450. https://doi.org/10.1007/s11528-018-0325-2.

McElheron, P., and M. Pilgaard Harsaae. 2016. BETTER INNOVATION BY DESIGN? CAN A COLLABORATIVE CROSS-DISCIPLINARY APPROACH ASSIST A PARADIGM SHIFT IN EDUCATION PRACTICE? DS 83: Proceedings of the 18th International Conference on Engineering and Product Design Education (E&PDE16), Design Education: Collaboration and Cross-Disciplinarity, Aalborg, Denmark, 8th-9th September 2016.

McKilligan, S., N. Fila, D. Rover, and M. Mina. 2017. "Design Thinking as a Catalyst for Changing Teaching and Learning Practices in Engineering." 2017 IEEE Frontiers in Education Conference (FIE).

McLaughlin, J. E., M. D. Wolcott, D. Hubbard, K. Umstead, and T. R. Rider. 2019. "A Qualitative Review of the Design Thinking Framework in Health Professions Education." *BMC Medical Education* 19:1–8.

Meiksins, P., P. Layne, E. Camargo, and K. Snead. 2013. "Women in Engineering: A Review of the 2013 Literature." *SWE Magazine* 60: 38–56.

Melles, G., Z. Howard, and S. Thompson-Whiteside. 2012. "Teaching Design Thinking: Expanding Horizons in Design Education." *Procedia-Social and Behavioral Sciences* 31:162–166. https://doi.org/10.1016/j.sbspro.2011.12.035.

Mentzer, N., B. Krishna, A. Kotangale, and L. Mohandas. 2023. "HyFlex Environment: Addressing Students' Basic Psychological Needs [Article]." *Learning Environments Research* 26 (1): 271–289. https://doi.org/10.1007/s10984-022-09431-z.

Micheli, P., S. J. Wilner, S. H. Bhatti, M. Mura, and M. B. Beverland. 2019. "Doing Design Thinking: Conceptual Review, Synthesis, and Research Agenda." *Journal of Product Innovation Management* 36 (2): 124–148. https://doi.org/10.1111/jpim.12466.

Mikic, B., and A. Rudnitsky. 2016. "An Emergent Approach to Design Thinking and Collaborative Team Composition in an Introduction to Engineering Course [Article; Proceedings Paper]." *International Journal of Engineering Education* 32 (3): 1514–1521. <Go to ISI>://WOS:000389146200022.

Mills, J. E., and D. F. Treagust. 2003. "Engineering Education – Is Problem-Based or Project-Based Learning the Answer." *Australasian Journal of Engineering Education* 3 (2): 2–16.

Mitcham, C. 1994. *Thinking through Technology: The Path between Engineering and Philosophy*. Chicago: University of Chicago Press.

Moher, D., A. Liberati, J. Tetzlaff, D. G. Altman, and PRISMA Group*, t. 2009. "Preferred Reporting Items for Systematic Reviews and Meta-analyses: The PRISMA Statement." *Annals of Internal Medicine* 151 (4): 264–269. https://doi.org/10.7326/0003-4819-151-4-200908180-00135.

Nicholas, D., A. Watkinson, H. R. Jamali, E. Herman, C. Tenopir, R. Volentine, S. Allard, and K. Levine. 2015. "Peer Review: Still King in the Digital age." *Learned Publishing* 28 (1): 15–21. https://doi.org/10.1087/20150104.

Nilsen, P. 2020. "Making Sense of Implementation Theories, Models, and Frameworks." In *Implementation Science 3.0*, edited by B. Albers, A. Shlonsky, and R. Mildon, 53–79. Cham: Springer.

North, C. E. 2008. "What Is All This Talk about "Social Justice"? Mapping the Terrain of Education's Latest Catchphrase." *Teachers College Record* 110 (6): 1182–1206. https://doi.org/10.1177/016146810811000607.

Nussbaum, M. 2004. "Emotions as Judgments of Value and Importance." In *Thinking about Feeling: Contemporary Philosophers on Emotions*, edited by R. C. Solomon, 183–199. Oxford: Oxford University Press.

O'Connor, S., J. Power, and N. Blom. 2023. "A Systematic Review of CDIO Knowledge Library Publications (2010–2020): An Overview of Trends and Recommendations for Future Research." *Australasian Journal of Engineering Education* 28 (2): 166–180. https://doi.org/10.1080/22054952.2023.2220265.

Ostrowski, S., R. Rolczyński, J. Pniewska, and I. Garnik. 2015. "User-friendly e-Learning Platform: A Case Study of a Design Thinking Approach use." Proceedings of the Mulitimedia, Interaction, Design and Innnovation, Warsaw, Poland. https://doi.org/10.1145/2814464.2814483.

Owen, C. 2007. "Design Thinking: Notes on Its Nature and use." *Design research quarterly* 2 (1): 16–27.

Oxman, R. 2001. "The Mind in Design: A Conceptual Framework for Cognition in Design Education." In *Design Knowing and Learning: Cognition in Design Education*, edited by C. Eastman, W. Newstetter, and M. McCracken, 269–295. Amsterdam: Elsevier.

Palacin-Silva, M., J. Khakurel, A. Happonen, T. Hynninen, and J. Porras. 2017, Nov 07-09. "Infusing Design Thinking Into a Software Engineering Capstone Course." *Conference on Software Engineering Education and Training* [2017 Ieee 30th Conference on Software Engineering Education and Training (csee&t)]. 30th IEEE Conference on Software Engineering Education and Training (CSEE&T), Savannah, GEORGIA.

Panke, S. 2019. "Design Thinking in Education: Perspectives, Opportunities and Challenges." *Open Education Studies* 1 (1): 281–306. https://doi.org/10.1515/edu-2019-0022.

Papaioannou, D., A. Sutton, and A. Booth. 2016. *Systematic Approaches to a Successful Literature Review*. 2nd ed. London: Sage Publications.

Petticrew, M., and H. Roberts. 2008. *Systematic Reviews in the Social Sciences: A Practical Guide*. Oxford: John Wiley & Sons.

Phillips, M., J. B. Reed, D. Zwicky, and A. S. Van Epps. 2023. "A Scoping Review of Engineering Education Systematic Reviews." *Journal of Engineering Education* 113 (4): 818–837. https://doi.org/10.1002/jee.20549.

Ping, C. S., P. Chow, and C. Teoh. 2011. The use of Design Thinking in CDIO Projects. Proceedings of the 7th International CDIO Conference, Technical University of Denmark, Copenhagen.

Plattner, H. 2007. "Launching the Next Generation." *Newsweek* 150 (24): E6–E6.

Poursharif, G., T. P. Doss, R. Broadbent, and G. Knight. 2021. "Developing Global Engineers through Interdisciplinary PBL and Design Thinking." 2021 IEEE Global Engineering Education Conference (EDUCON), Vienna, AUSTRIAhttps://doi.org/10.1109/EDUCON46332.2021.9454051.

Powell, A., B. Bagilhole, and A. Dainty. 2009. "How Women Engineers Do and Undo Gender: Consequences for Gender Equality." *Gender, Work & Organization* 16 (4): 411–428. https://doi.org/10.1111/j.1468-0432.2008.00406.x.

Purzer, S., and J. P. Quintana-Cifuentes. 2019. "Integrating Engineering in K-12 Science Education: Spelling out the Pedagogical, Epistemological, and Methodological Arguments." *Disciplinary and Interdisciplinary Science Education Research* 1 (1). 13. https://doi.org/10.1186/s43031-019-0010-0.

Rahman, M. H., C. Schimpf, C. Xie, and Z. H. Sha. 2019. "A Computer-Aided Design Based Research Platform for Design Thinking Studies [Article]." *Journal of Mechanical Design* 141 (12): 121102. https://doi.org/10.1115/1.4044395.

Rak, D. 2023. Information Literacy in the Design Thinking Process-A Preliminary Research. European Conference on Information Literacy.

Ranger, B. J., and A. Mantzavinou. 2018. "Design Thinking in Development Engineering Education: A Case Study on Creating Prosthetic and Assistive Technologies for the Developing World." *Development Engineering* 3:166–174. https://doi.org/10.1016/j.deveng.2018.06.001.

Rao, R. Y., J. J. Koola, and N. D. Mehta. 2018, Dec 10-13. Design Engineering: A Curriculum on Design Thinking. *IEEE International Conference on Technology for Education* [2018 ieee Tenth International Conference on Technology for Education (t4e)]. IEEE Tenth International Conference on Technology for Education (T4E), Chennai, INDIA.

Rasoal, C., H. Danielsson, and T. Jungert. 2012. "Empathy among Students in Engineering Programmes." *European Journal of Engineering Education* 37 (5): 427–435. https://doi.org/10.1080/03043797.2012.708720.

Razzouk, R., and V. Shute. 2012. "What Is Design Thinking and why Is It Important?" *Review of Educational Research* 82 (3): 330–348. https://doi.org/10.3102/0034654312457429.

Reichenbach, K., B. Reynante, and Acm. 2019, Mar 09-10. Educating Change-Makers: A Framework for Sustainable Making and Activist Engineering. [Proceedings of 8th Annual Conference on Maker Education (Fablearn 2019)]. 8th Annual Conference on Maker Education (FabLearn), New York, NY.

Reynante, B. 2022. "Learning to Design for Social Justice in Community-Engaged Engineering." *Journal of Engineering Education* 111 (2): 338–356. https://doi.org/10.1002/jee.20444.

Ritchie, J., J. Lewis, C. M. Nicholls, and R. Ormston. 2003. *Qualitative Research Practice (Vol. 757)*. London: Sage.

Rodriguez, S. L., E. E. Doran, R. E. Friedensen, E. Martinez-Podolsky, and P. S. Hengesteg. 2020. "Inclusion & Marginalization: How Perceptions of Design Thinking Pedagogy Influence Computer, Electrical, and Software Engineering Identity [Article]." *International Journal of Education in Mathematics Science and Technology* 8 (4): 304–317. https://doi.org/10.46328/ijemst.v8i4.952.

Samuel, A., and W. Lewis. 1991. "Evaluation of Students' Strategies for Concept Application in Engineering Design." *Instructional Science* 20 (4): 311–329. https://doi.org/10.1007/BF00043256.

Sanders, M. E. 2008. Stem, Stem Education, Stemmania.

Saseendran, A., and M. V. Thomas. 2025. "Design Thinking in Science and Integrated STEM/STEAM Education: Trends, Challenges, and Future Directions from a Systematic Review." *STEM Education* 5 (6): 1058–1101. https://doi.org/10.3934/steme.2025046.

Scheer, A., C. Noweski, and C. Meinel. 2012. "Transforming Constructivist Learning into Action: Design Thinking in Education." *Design and Technology Education* 17 (3): 8–19.

Schön, D. A. 1987. Educating the Reflective Practitioner.

Schön, D. A. 1992. "Designing as Reflective Conversation with the Materials of a Design Situation." *Knowledge-Based Systems* 5 (1): 3–14. https://doi.org/10.1016/0950-7051(92)90020-G.

Schön, D. A. 2017. *The Reflective Practitioner: How Professionals Think in Action*. New York: Routledge.

Schreier, M., T. Dahl, M. Janssen, A. Whittal, and C. Stamann. 2020. "Qualitative Content Analysis: Disciplinary Perspectives and Relationships between Methods-Introduction to the FQS Special Issue" Qualitative Content

Analysis II." *Forum Qualitative Sozialforschung/Forum: Qualitative Social Research* 21 (1). Art. 1.https://doi.org/10.17169/fqs-21.1.3661.

Shah, J. J., S. M. Smith, and J. Woodward. 2009, Aug 24-27. "DEVELOPMENT OF STANDARDIZED TESTS FOR DESIGN SKILLS." [Iced 09 – the 17th International Conference on Engineering Design, vol 10: Design Education and Lifelong Learning]. 17th International Conference on Engineering Design, Stanford Univ, Stanford, CA.

Shetty, D., J. J. Xu, and Asme. (2018, Nov 09-15). STRATEGIES TO ADDRESS "DESIGN THINKING" IN ENGINEERING CURRICULUM. [Proceedings of the ASME International Mechanical Engineering Congress and Exposition, 2018, vol 5]. ASME International Mechanical Engineering Congress and Exposition (IMECE2018), Pittsburgh, PA.

Simon, H. A. 1973. "The Structure of ill Structured Problems." *Artificial Intelligence* 4 (3-4): 181–201. https://doi.org/10.1016/0004-3702(73)90011-8.

Simon, H. A. 2013. "The Scientist as Problem Solver." In *Complex Information Processing*, edited by D. Klahr and K. Kotovsky, 375–398. New York: Psychology Press.

Skolimowski, H. 1966. "The Structure of Thinking in Technology." *Technology and Culture* 7 (3): 371–383. https://doi.org/10.2307/3101935.

Somerville, M., D. Anderson, H. Berbeco, J. R. Bourne, J. Crisman, D. Dabby, H. Donis-Keller, S. S. Holt, S. Kerns, and D. V. Kerns. 2005. "The Olin Curriculum: Thinking toward the Future." *IEEE Transactions on Education* 48 (1): 198–205. https://doi.org/10.1109/TE.2004.842905.

Somerville, M., J. Townsend, and Ieee. 2014, Oct 22-25. A Student-Centered Approach to Designing Teaming Experiences.*Frontiers in Education Conference* [2014 ieee Frontiers in Education Conference (fie)]. IEEE Frontiers in Education Conference (FIE), Madrid, SPAIN.

Soomro, S. A., H. Casakin, and G. V. Georgiev. 2021. "Sustainable Design and Prototyping Using Digital Fabrication Tools for Education [Article]." *Sustainability* 13 (3). 1196https://doi.org/10.3390/su13031196.

Souza, A., B. Ferreira, N. Valentim, L. Correa, S. Marczak, and T. Conte. 2020. "Supporting the Teaching of Design Thinking Techniques for Requirements Elicitation through a Recommendation Tool [Article]." *Iet Software* 14 (6): 693–701. https://doi.org/10.1049/iet-sen.2019.0300.

Srisawat, S., and P. Wannapiroon. 2022. "The Development of Virtual Professional Learning Community Platform with Experiential Design Thinking Process to Enhance Digital Teacher Competency." *International Journal of Information and Education Technology* 12 (12): 1196. https://doi.org/10.18178/ijiet.2022.12.12.1753.

Steuer-Dankert, L., S. K. Gilmartin, C. B. Muller, C. Dungs, S. Sheppard, and C. Leicht-Scholten. 2019. "Expanding Engineering Limits-A Concept for Socially Responsible Education of Engineers [Article]." *International Journal of Engineering Education* 35 (2): 658–673. <Go to ISI>://WOS:000459427000019.

Stufflebeam, D. L. 2003. "Institutionalizing Evaluation in Schools." In *International Handbook of Educational Evaluation*, edited by T. Kellaghan and D. L. Stufflebeam, 775–805. Dordrecht: Springer.

Sukackė, V., A. O. P. d. C Guerra, D. Ellinger, V. Carlos, S. Petronienė, L. Gaižiūnienė, S. Blanch, A. Marbà-Tallada, and A. Brose. 2022. "Towards Active Evidence-Based Learning in Engineering Education: A Systematic Literature Review of PBL, PjBL, and CBL." *Sustainability* 14 (21): 13955. https://doi.org/10.3390/su142113955.

Suligoj, V., R. Zavbi, and S. Avsec. 2020. "Interdisciplinary Critical and Design Thinking [Article]." *International Journal of Engineering Education* 36 (1): 84–95. <Go to ISI>://WOS:000506204800007.

Thomas, J., X. Chen, I. Lee, J. Wang, Y. Yang, and M. Strickfaden. 2021. Learning Design Thinking Through a Collaborative Focus on Social Justice. Advances in Industrial Design: Proceedings of the AHFE 2021 Virtual Conferences on Design for Inclusion, Affective and Pleasurable Design, Interdisciplinary Practice in Industrial Design, Kansei Engineering, and Human Factors for Apparel and Textile Engineering, July 25-29, 2021, USA.

Tiwari, S., and S. S. Rathore. 2022. "Understanding General Concepts of Requirements Engineering through Design Thinking: An Experimental Study with Students [Article]." *Computer Applications in Engineering Education* 30 (6): 1683–1700. https://doi.org/10.1002/cae.22550.

Todoroff, E. C., T. Shealy, J. Milovanovic, A. Godwin, and F. Paige. 2021. "Comparing Design Thinking Traits between National Samples of Civil Engineering and Architecture Students." *Journal of Civil Engineering Education* 147 (2): 04020018. https://doi.org/10.1061/(ASCE)EI.2643-9115.0000037.

Ulibarri, N., A. E. Cravens, M. Cornelius, A. Royalty, and A. S. Nabergoj. 2014. "Research as Design: Developing Creative Confidence in Doctoral Students through Design Thinking." *International Journal of Doctoral Studies* 9:249–270. https://doi.org/10.28945/2062.

Vaismoradi, M., and S. Snelgrove. 2019. Theme in Qualitative Content Analysis and Thematic Analysis.

van den Akker, J. J., W. Kuiper, and U. Hameyer. 2003. *Curriculum Landscapes and Trends*. Dordrecht: Springer.

Van Helden, G., B. T. Zandbergen, M. M. Specht, and E. K. Gill. 2023. "Collaborative Learning in Engineering Design Education: A Systematic Literature Review." *IEEE Transactions on Education* 66 (5): 509–521. https://doi.org/10.1109/TE.2023.3283609.

Von Thienen, J. P., W. J. Clancey, G. E. Corazza, and C. Meinel. 2018. "Theoretical Foundations of Design Thinking: Part I: John E. Arnold's Creative Thinking Theories." In *Design Thinking Research: Making Distinctions: Collaboration versus Cooperation*, edited by H. Plattner, C. Meinel, and L. Leifer, 13–40. Cham: Springer.

Wolmarans, N. 2016. "Inferential Reasoning in Design: Relations between Material Product and Specialised Disciplinary Knowledge." *Design Studies* 45:92–115. https://doi.org/10.1016/j.destud.2015.12.003.

Wu, B., Y. Hu, and M. Wang. 2019. "Scaffolding Design Thinking in Online STEM Preservice Teacher Training." *British Journal of Educational Technology* 50 (5): 2271–2287. https://doi.org/10.1111/bjet.12873.

Yakub, B., S. M. Naik, and A. Unnisa. 2018. "Design Thinking Implementation by Faculty through EPICS Project." *Journal of Engineering Education Transformations* 32 (2): 7–10. https://doi.org/10.16920/jeet/2018/v32i2/139490.

Yu, Q., K. Yu, and R. Lin. 2024. "A Meta-analysis of the Effects of Design Thinking on Student Learning." *Humanities and Social Sciences Communications* 11 (1): 1–12. https://doi.org/10.1057/s41599-024-03237-5.

Zancul, E. d. S, L. F. dos S Durao, R. d. D Lopes, D. Nakano, P. Blikstein, G. G. Majzoub, and D. L. Dalmon. 2017. "An Empirical Study on Design-Based vs. Traditional Approaches in Capstone Courses in Engineering Education." *International Journal of Engineering Education* 33 (5, SI): 1543.